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Hospital Standard - A study case for interoperability in health care sector

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Abstract

This report presents a case study on interoperability in the Swedish healthcare sector, focusing on the integration of the clinical process visualization tool ActMap with standardized health information models, particularly openEHR. The project adopts a mixed-methods approach that combines technical implementation with human-centred research in order to evaluate both technical feasibility and practical relevance in real clinical contexts.

From a technical perspective, two automated terminology mapping approaches were designed and evaluated. The first is a rule-based translator based on string similarity algorithms, which demonstrated high transparency and computational efficiency, successfully mapping approximately 90% of ActMap terms to openEHR archetypes, although the majority of mappings required expert validation. The second approach leverages Large Language Models (LLMs) to perform semantic mapping. Despite improved flexibility, the LLM-based method achieved limited accuracy (approximately 44%), revealing challenges related to small training datasets, semantic ambiguity, and contextual asymmetry between local clinical workflows and standardized archetype structures.

In parallel, qualitative and quantitative data were collected through interviews and surveys with healthcare professionals. The findings highlight persistent interoperability issues in daily clinical practice, including fragmented systems, extensive documentation requirements, and a usability paradox in which standards intended to streamline workflows instead increase cognitive and administrative burden. Clinicians expressed strong demand for intuitive interfaces, automated data entry, and interoperability mechanisms that operate seamlessly in the background without disrupting patient care.

The study concludes that while automated mapping tools offer a viable pathway toward improved interoperability, human expertise remains essential for validation and contextual accuracy. The results emphasize the importance of aligning technical solutions with real clinical needs and workflows, supporting the adoption of standards such as openEHR and compliance with emerging regulations like the European Health Data Space (EHDS). This course project contributes to a proof-of-concept (PoC) for bridging hospital-specific systems with standardized health information models, aiming to enhance data reusability, reduce administrative overhead, and support safer and more efficient healthcare delivery.

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Acronyms

AI Artificial Intelligence. 8, 13

CKM Clinical Knowledge Manager. 19

CNAS Romanian's National Health Insurance House. 35

CSV Comma-Separated values. 20

DICOM Digital Imaging and Communications in Medicine. 7, 9

EHDS European Health Data Space. 1, 13, 14, 47

EHR Electronic Health Record. 7, 8, 11, 14

ETL Extracting, Transforming, and Loading. 15

FHIR Fast Healthcare Interoperability Resources. 14, 16

GDPR General Data Protection Regulation. 13, 14, 47

HIS Hospital Information System. 35

HL7 Health Level Seven. 7, 8, 14

IT Information & Technology. 7–10, 12–14

LIS Laboratory Information System. 35

LLM Large-Language Model. 16

PACS Picture Archiving and Communication System. 9

PDL Swedish Patient Data Act. 13, 14

RIS Radiology Information System. 8

SI structured interviews. 17

SSI semi-structured interviews. 17

UI User Interface. 10

Glossary

ActMap An application to simply the interoperability between medical process in healthcare systems. 7–10, 12–16

JSON An open standard file format and data interchange format that uses human-readable text to store and transmit data objects. 15, 16

openEHR An open, vendor-neutral standard for structuring and managing electronic health record applications. 7, 11–16

XML An markup language and file format for storing, transmitting, and reconstructing data. 15

1 Introduction

Healthcare systems increasingly rely on complex information technologies to document, manage and exchange clinical data. While digitalization has enabled improved access to patient information and advanced medical workflows, it has also introduced significant challenges related to interoperability, usability and administrative burden. Clinicians must navigate fragmented Information & Technology (IT) landscapes containing multiple standards, systems and interfaces. This results in inconsistent data, poor cooperation, duplicated work and ultimately reduced time available for actual patient care.

Within the Swedish healthcare system, these challenges are particularly evident due to the coexistence of several Electronic Health Record (EHR) systems and data standards, such as Health Level Seven (HL7), Digital Imaging and Communications in Medicine (DICOM) and often local department-specific adapted solutions. On paper, some of these standards should be interoperable. However, between their parallel use and varying implementations across regions and hospitals, their effectiveness is reduced in practice. As a result, valuable clinical information may be difficult to locate, interpret, or reuse across organisational boundaries.

This course project was conducted by students at Uppsala University for the advanced-level course `IT & Society`. In collaboration with the company data2value and Akademiska Sjukhuset, the project focuses on ActMap, an application aimed at improving process visualisation and information structuring within clinical workflows. ActMap seeks to empower healthcare professionals by making relevant information accessible at the right time and in a form that supports daily clinical work.

The primary two objectives of this project is to explore how ActMap can be connected to standardized health information models, with a particular focus on openEHR. By investigating both rule-based methods and large language model–based approaches for mapping ActMap workflow elements to openEHR archetypes, the project evaluates the technical feasibility, limitations, and potential benefits of automated terminology translation. The second objective was to assess the practical relevance and perceived value of such a solution in real clinical settings. This was done using qualitative and quantitative methods, conducting interviews and surveys with medical

professionals.

Through this combined technical and human-centred approach (also resembled in the class' structure, all students divided in two small teams, each having a general manager, a report manager and members), the project aims to contribute towards how standardised data models and emerging Artificial Intelligence (AI) techniques improve interoperability and the usability of digital health systems, along with a look into what the current state might look like.

2 Background

This section will focus on introducing the goal and purpose of the course project, understanding the client's need, along some preliminary study/research on standards useful for interoperability in healthcare sector.

2.1 ActMap - Goal and Purpose

To provide sufficient context for understanding the goals and intended use of ActMap, this subsection first presents a brief overview of the Swedish healthcare IT infrastructure. This is followed by an introduction to ActMap itself, outlining its purpose, guiding principles, and role within the broader clinical information ecosystem.

2.1.1 Health Care Overview

Before having an overview about what this ActMap application is trying to achieve, it is important to understand how the Information & Technology (IT) infrastructure is done within the Swedish Health System.

As a simple use case, you go to a central health point and give explain some of the symptoms. You are then recorded into an Electronic Health Record (EHR) application (e.g. Cosmic) which is going to have a dedicated page with all of the present, past and future medical records that are related to you. All of this information then goes to Radiology Information System (RIS), using the Health Level Seven (HL7) standards. After that, into to a huge server, taking numerous paths/branches with information, containing

different standards. Furthermore, it can arrive to multiple storage units, to other servers to communicate with other hospitals and to other applications where clinicians are able to visualize the data. The main communication protocol is Digital Imaging and Communications in Medicine (DICOM)[14, 20]. One of these important storage units is Picture Archiving and Communication System (PACS) which is the hospital's system used to store and view medical images. An overview visualisation of this description can be seen in Figure 1.

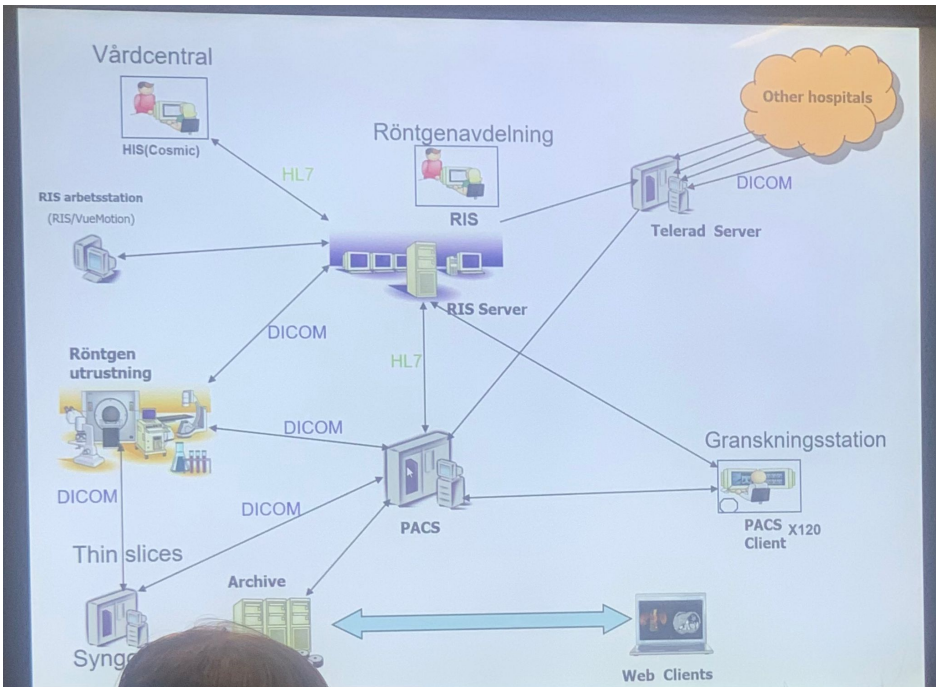


Figure 1: Overview of the whole IT infrastructure at the Hospital level including PACS systems [1]

2.1.2 ActMap Overview

The startup company data2value is exploring ideas for developing meaningful integrations within the Swedish healthcare sector. At the core of this initiative lies ActMap, a solution designed to enhance process mapping and improve the management of clinical data. We will present several paragraphs taken directly from an initial presentation shared by one of their IT consul-

tants, Markus Eriksson.

"Our ambition is to improve healthcare – in Sweden, developing countries and globally - through empowering the clinical personnel in reaching and utilizing the right information and data, at the right time and in such a way that it fits and supports the clinical work. Our solutions are specifically designed to improve the clinical conditions to:

1. improve patient safety and support person- centered care
2. achieve a high quality and equal healthcare
3. bring a sense of calm and control in the organization
4. implement and support a continuous data driven improvement work"[4]

So, how can an application achieve these? By continuing to quote: "What is ActMap? An innovation for information structuring and user centric digitalization. In ActMap, vital information and data are connected. The potential of the existing system landscape is maximized while new efficient development opportunities emerge. ActMap brings together and makes information and data accessible, customized to the unique needs of clinical practice.".[4] Which is, in simpler terms, the perfect tool for a clinician when performing a medical procedure, easily reachable within the application, clear visualisation at any time through an user-friendly User Interface (UI) design integration. This can be view as the purpose of this application and the goal is to improve the healthcare sector with innovative IT solutions.

As presented and emphasized during a subsequent demonstration session, Uppsala University Hospital (Akademiska Sjukhuset) serves as the pioneering foundation and the primary pilot case for this application. Its involvement provides the opportunity to gather insights from real users within a complex, large-scale healthcare environment. The hospital, located in one of Sweden's most prosperous regions, employs a diverse workforce, utilizes advanced medical equipment, and encompasses numerous departments that must continuously communicate and collaborate effectively to ensure a shared understanding across different events.

[1]Picture taken from one study visit during the course *Medical Informatics - 1MD026/1MD030, 5hp* here at the Uppsala University

2.2 The openEHR Standard: A Foundation for Lifelong Health Records

This subsection introduces the openEHR standard, which serves as a key reference framework for semantic interoperability in healthcare information systems. It first explains the fundamental concepts and design principles behind openEHR, then outlines how the standard operates in practice, and finally discusses its relevance to the ActMap project and the motivations for exploring it as a potential integration target.

2.2.1 What is openEHR?

OpenEHR is an open, vendor-neutral standard for structuring and managing EHRs [3]. Its core innovation is a two-level modelling approach that fundamentally separates the definition of clinical concepts from the actual patient data. This architecture is designed to create flexible, interoperable, and long-lasting health information systems.

openEHR has gained widespread adoption in healthcare systems across multiple countries and regions including Europe (particularly the United Kingdom, Sweden, and the Netherlands), Australia, and Japan. It was accepted by the International Organisation for Standardisation in 2008 and developed into the ISO 13606-2 standard which was recently reviewed in 2019 and confirmed in 2024 [9]. To date, national electronic health record data centres in over 15 countries have adopted this standard, including notable implementations in the United Kingdom's National Health Service (NHS), Slovenia's national eHealth system, and Australia's My Health Record system.

According to its foundational principles, openEHR aims to [10]:
Promote and publish the formal specification of requirements for representing and communicating EHR information, based on implementation experience, and evolving over time as health care and medical knowledge develop;
Promote and publish EHR information architectures, models and data dictionaries, tested in implementations, which meet these requirements;
Maintain open source "reference" implementations, to enhance the pool of available tools to support clinical systems.

2.2.2 How does it work?

The openEHR methodology is built on two key layers:
The Archetype & Template Layer (Knowledge Layer): This layer is driven by clinical experts, not just IT personnel. Archetypes are reusable, granular data models that define specific clinical concepts such as "blood pressure," "laboratory result," or "allergy." They are built to be universally understood and are managed in shared international repositories. Templates then assemble these archetypes into meaningful, context-specific forms for use in clinical practice, such as a "Diabetes Consultation" form.

The Data Layer (Instance Layer): This layer contains the actual patient data. Every clinical entry (e.g., a specific blood pressure reading for a patient) is stored as an instance that conforms to the structure of an archetype. This ensures that the data is semantically consistent and computable.

2.2.3 Why is it relevant?

The students of this course were divided into different work groups and, for the duration of the first period, we researched a list of different medical standards that are widely used and different hospitals work with a combination of them. We therefore narrowed our focus to standards most relevant to ActMap's operational context, with the potential to help transform ActMap from an aggregation of existing standards into a cohesive, standardized system in its own right. We also thought that openEHR might be a good standard to use, at a certain level, within ActMap.

OpenEHR is relevant to the ActMap project and the broader Swedish healthcare context for several reasons, and our technical team's exploration of an AI-powered Translator to bridge ActMap's interface with the openEHR ecosystem serves as a concrete and strategic example of this relevance. **Semantic Interoperability as a Service:** The core problem openEHR solves is semantic interoperability – ensuring data means the same thing across systems. Our proposed Translator is designed to directly operationalize this principle. Instead of ActMap needing to be rebuilt as a native openEHR system, the Translator can intelligently map and transform data from ActMap's internal model into standardized, computable openEHR archetypes. This would effectively position ActMap as a gateway to the openEHR world, en-

abling seamless data exchange with other systems that adhere to this robust standard, without sacrificing ActMap's existing architecture.

Enabling Advanced Analytics and Innovation: A unified, high-quality data pool is the foundation for Artificial Intelligence (AI) and advanced analytics. By channelling diverse data sources through the Translator into a consistent openEHR format, ActMap can create a clean, structured data asset. This directly enables downstream innovation, such as the development of reliable clinical decision support systems, large-scale research, and population health analytics, thereby increasing the platform's overall value proposition.

2.3 The Policy Landscape for Health Data in Sweden

The development and implementation of health IT solutions in Sweden, including a platform like ActMap, are fundamentally shaped by a stringent and multi-layered policy framework. Compliance with these regulations is not optional but a prerequisite for market entry and operational legitimacy. Key pillars of this framework include the General Data Protection Regulation (GDPR), the emerging European Health Data Space (EHDS) regulation, and the national Swedish Patient Data Act (PDL).

2.3.1 General Data Protection Regulation

GDPR is the cornerstone of personal data protection across the European Union (EU), including Sweden [5]. For health data, which is classified as a "special category of data" under Article 9, the regulation imposes the highest standards of security and privacy. Its core principles, such as lawfulness, fairness and transparency, purpose limitation, data minimisation, and integrity and confidentiality, directly dictate how systems like ActMap must be designed and operated.
This means that from the initial architecture phase, ActMap must embed 'Data Protection by Design and by Default'. Technically, this translates to implementing robust encryption for data at rest and in transit, stringent access control mechanisms, and comprehensive audit trails to log every access to patient data. Furthermore, GDPR grants patients (data subjects) rights such as the right to access, rectify, and even erase their data, which the platform must facilitate. Non-compliance can result in severe financial penalties

of up to 4% of global annual turnover, making GDPR a critical consideration for any digital health venture [22].

2.3.2 European Health Data Space

EHDS regulation represents a paradigm shift, aiming to create a single market for health data across the EU [8]. While GDPR focuses on protection, the EHDS actively promotes the secure and authorized secondary use of health data for research, innovation, and policy-making. For ActMap, the EHDS presents both a future compliance obligation and a significant strategic opportunity.
The regulation will likely mandate the use of specific technical standards for EHRs and data exchange to ensure interoperability. While the final list is under development, standards like openEHR and HL7 Fast Healthcare Interoperability Resources (FHIR) are strong candidates given their alignment with the EHDS's goals. By proactively designing ActMap's architecture to be compatible with these standards, the platform can position itself as a future-proof solution ready to leverage the cross-border data access and large-scale data pooling opportunities that the EHDS will enable. Early alignment is not merely about compliance but about gaining a first-mover advantage in the emerging European health data economy.

2.3.3 The Swedish Patient Data Act (Patientdatalagen, PDL)

Complementing and specifying GDPR at the national level is the Swedish Patient Data Act (PDL). The PDL provides the concrete legal framework for the handling of all patient records within the Swedish healthcare system. It is a literal gatekeeper for any integration with Swedish healthcare providers. A core principle of the PDL is the concept of "journalskydd" (record protection), which strictly regulates which healthcare professionals can access a patient's record. Any system, including ActMap, that is to be integrated into a clinical workflow must demonstrate an unparalleled level of access control, ensuring that only personnel directly involved in a patient's care can view their data. Furthermore, the PDL solidifies the patient's right to block cross-access ("blockad") to their record, a feature that must be strictly respected by any integrated IT system. Therefore, the feasibility of ActMap's features in a Swedish context is entirely dependent on its ability to meet the concrete, legally binding criteria set forth in the PDL.

3 Related Work

3.1 Process Visualization Techniques

Research by Pini et al. (2015) in process modelling and process mining shows that good visualizations are important for understanding complex workflows, especially in healthcare where processes often have to be followed in a specific order for the best interest of the patient. The research paper by Pini et al. (2015) have shown that visual diagrams can make it easier to see patterns and differences in how processes are carried out compared to only text or tables. Their research also highlight that clear and accessible visualizations are important for supporting analysts, since tabular data alone can be hard to interpret in complex healthcare workflows [17]. This shows a general need for tools that make process information easier to understand and work with, especially in clinical environments where multiple systems and data formats exist.

3.2 OpenEHR Mapper

Reimer et al. (2012) present the openEHR Mapper, a tool that is designed to translate clinical and genomic data into openEHR structures so they can be used in their systems. Their approach is based on mapping input data fields to openEHR archetypes using lookup tables and then generating valid openEHR compositions. The tool processes JSON or XML inputs, builds openEHR objects and uploads them to an openEHR server. One important point in their work is that they show how structured data from different sources can be converted into a standardised openEHR format through relatively simple mappings [19]. This is relevant for the project as the goal is to create a translator that takes ActMap-style JSON files and converts selected fields into an openEHR-like structure.

A similar focus on data transformation appears in the work by Tute and Steiner (2018), who describe how clinical data warehouses rely on Extracting, Transforming, and Loading (ETL) processes from different hospital systems. They argue that ETL steps are often complex and poorly documented, highlighting how standardized models and clear mappings can make data integration easier and more reliable. The study shows that standardized archetypes are important for transforming data between different systems [21]. This is

a relevant topic valuable to take into consideration when implementing the translator that converts ActMap JSON files to openEHR-like structure.

The study conducted by Lads et al. (2022) presented a practical implementation of a conversion pipeline between openEHR and Fast Healthcare Interoperability Resources (FHIR) as part of the German Corona Consensus Dataset. The work shows how structured clinical content modelled in openEHR compositions is transformed into FHIR resources. There are several technical challenges that occurred for bridging standards, such as semantic mismatches, differences in granularity and the need for careful handling of terminology bindings. The authors raise the importance of maintaining data integrity and source origin during conversion. The paper illustrates the complexities associated with mapping between standards and shows how a translation pipeline must account for both structural and semantic consistency [12], which should be taken into account for this project as well.

3.3 LLMs and Schema Matching

There is some research examining whether Artificial Intelligence can support or automate parts of mapping processes. The experimental study by Parciak et al. (2024) evaluates the effectiveness of a Large language Model for schema matching between two relational schemas based only on the attribute names and textual descriptions. Their work focuses on scenarios where only metadata is available. The paper shows that match quality decreases when there is too little or too much context provided and that newer LLM versions improve decisiveness. The authors conclude that LLMs can accelerate parts of the schema matching process. This study is relevant to this project, using AI for mapping between ActMap files and openEHR, because it is about aligning structurally different data models. It is demonstrated that LLMs can perform semantic matching using only metadata [16], similar conditions to which ActMap and openEHR mapping would need.

4 Methodology

This chapter presents an overview of the methodological approaches employed in our course project. The methodology employs both qualitative and quantitative methods, including surveys and interviews, with corresponding

analytical techniques such as content analysis and inferential statistics to interpret the sampled data. In addition to these human-science-oriented approaches, technical methodology is adapted to the project. The technical approach involves selecting an appropriate large language model and defining evaluation methods to interpret the results from the dataset.

4.1 Interview methodology

This approach, which includes conducting interviews, was chosen to better understand how different medical professionals apply standards in their daily work and to assess whether a tool that consolidates these elements would be useful for them. The aim is to gain a more in-depth perspective on digitalisation and to evaluate whether such a tool, with interoperability capabilities, could simplify and improve their workflows.

In the process of preparing for the interviews, two types of interview methods were discussed: Semi-structured interviews (SSI) and Structured interviews (SI). Before deciding on a specific method, the main questions were written along with additional follow-up questions. During this process, uncertainty on interviewees' knowledge arose, resulting in a focus on SI interviews. One benefit with SSI is that it shifts the focus to quality instead of quantity and gives the interviewer a chance to come up with new and relevant questions during the interview to capture more data [1]. A disadvantage of SSI is that it is time-consuming, which was an important factor during this project [1]. When reaching out to different medical staff, the decision of using SSI would be too time-consuming and SI was the right method for this project. Despite this, one SSI interview was conducted, which was about 20 minutes long and was conducted in Swedish. If you combine the SSI with the rest of the interview, a total of five (5) interviews. Using the SI method, the questions were sent to the people being interviewed, and they sent back their answers. The interviewees consist of both Swedish and international medical staff, thus the interviews were done in either Swedish, English, German or Romanian. An LMM chatbot, such as Copilot or ChatGPT, was used to translate both the questions and answers from the native languages (except the English one) into English. This is stated in the Appendix B, where also the full transcript (non- and translated versions) of the interviews are available.

Content analysis and thematic analysis are qualitative approaches that treat

interview transcripts as textual data from which meaning can be systematically derived. Content analysis emphasizes systematic coding, categorization, and often counting of quoting some elements from the text to better describe patterns and sometimes link them to a broader existing framework or phenomena. Thematic analysis focuses on identifying, analysing, and interpreting recurrent patterns of meaning (themes) across the data, and is theoretically flexible, allowing researchers to work within different codes and/or themes to understand participants' experiences and perspectives. Both methods are interconnected, and can be used along each other to better emphasise recurring patters with different coding styles. [15]

4.2 Survey methodology

Since the interviews focused on qualitative data, more in-depth analysis potential and questioning, the purpose of the survey is to get a larger quantity of answers based on similar needs, mostly to be related on the practicability of such an application/tool and how would impact the medical personnel. Therefore, the survey is written with a focus on being easy to answer and having a short duration. This design minimizes the effort and time from the participant and increases completion rates. The questions are a mix between closed-ended questions and open-ended question, including multiple-choice and Likert scale (questions with answers based on a scale - e.g. 1 to 5 or from the worst to the best and everything in between) [2] for statistical analysis, and text fields for more in-depth answers. The survey was distributed via direct contact to different hospitals in Sweden and to medical staff, using the snowball effect [7], in diverse areas, ensuring a broad sample.

Descriptive and inferential statistical analyses are quantitative approaches that conceptualize survey data as observations from a sample drawn from a population, grounded in probability theory. Descriptive statistics summarize and organize the observed data using measures such as dispersion, graphical displays etc., providing a theoretical basis for characterising the sample without extending claims beyond it. Inferential statistics use sampling distributions and probability models to estimate population parameters and test hypotheses, allowing researchers to make theoretically justified generalisations and predictions about the wider population from which the survey sample is drawn. The great fit for our project will be to choose the inferential analysis over a mixture between the two of them because we want to have

predictions and make generalised assumption based on the analysed data, not to stay only in the scope of the sample without going beyond it. [11]

4.3 Dataset

Markus provided a dataset containing ActMap JSON headers. These headers were compared to openEHR archetypes with a Python script to get an overview of the compatibility between the two standards. This is explained in further details in Section 4.4. These headers were also manually mapped to corresponding openEHR archetypes, resulting in a complete dataset in which 26 of the original JSON headers were successfully aligned with appropriate archetypes. However, this dataset was far too small to be viable for training an AI model. Therefore, generative AI was used to synthesize realistic data in order to increase the dataset size. Each example included an ActMap node name, a list of related links containing one correct Clinical Knowledge Manager (CKM) archetype link and several plausible but incorrect alternatives, and a target JSON object specifying the correct openEHR entry type and archetype identifier.

4.4 Evaluation Metrics and Similarity Measurement

This subsection describes the methodology behind the Python script mapping openEHR archetypes to ActMap's JSON headers. Details of the modules mentioned in this section can be found in the repository in Appendix A.

4.4.1 Rule-based Translator's Similarity Measurement

The system employs a string edit distance–based similarity algorithm as the primary metric for evaluating mapping quality. This approach is implemented using the `SequenceMatcher` algorithm from Python's standard `difflib` library, which measures the similarity between two strings by estimating the minimum number of edit operations required to transform one string into the other. The resulting value is normalized to produce a similarity score ranging from 0.0 to 1.0, where higher values indicate greater similarity.

The similarity computation is integrated into two main processing modules. The first module, `extract.py`, calculates the similarity between each

ActMap term and all available openEHR archetypes, selecting the archetype with the highest similarity score as the best match. The second module, `reverse.py`, performs the inverse operation by identifying, for each openEHR archetype, the most similar ActMap term based on the same similarity.

The system outputs two Comma-Separated values (CSV) files (`actmap_to_openehr_mapping.csv` and `openehr_to_actmap_mapping.csv`), both containing the similarity field that records the similarity score for each mapping pair. Based on the observed mapping results, a set of quality thresholds was established to classify the confidence of the generated mappings. Similarity scores of ≥ 0.7 were considered to indicate high confidence mappings and could be directly adopted without further review. Scores in the range of 0.4 to 0.7 were classified as medium confidence, for which manual inspection was recommended. Mappings with similarity scores below 0.4 were categorized as low confidence and are not considered.

This method offers the advantages of computational efficiency, transparent results, and simple implementation, providing a quantifiable evaluation baseline for automated terminology mapping. The main limitation lies in its consideration of only literal similarity, lacking understanding of semantic relationships in medical terminology and clinical context.
This similarity measurement method provides a foundational reference for subsequent performance comparisons with LLM-based approaches.

4.5 LLM Model Selection and Training

The goal for the part of the project regarding Large Language Models (LLM) was to evaluate whether LLMs can be used to map semi-structured ActMap workflow nodes to structured openEHR archetypes. The task requires both instruction-following and structured reasoning, while producing strictly formatted JSON output. For this reason, we focused on instruction-tuned, open-weight models that are realistic to fine-tune and deploy in practice. A small and local LLM that can run on 8–16 GB RAM memory, perform semantic mapping between ActMap process tags and openEHR terminology to produce stable JSON mapping files is required. Because this is a low-resource and mapping-only workload, compact 3–7B models are ideal.

Phi-3 Mini (4k instruct)[6], Qwen 2.5 (3B)[18], and Mistral 7B[13] were the

selected models. These models range between 3–7 billion parameters, which was a deliberate choice. Models of this size offer a balance between reasoning capability and computational feasibility, especially in enterprise or healthcare settings where very large models may be impractical due to cost, and latency.

Phi-3 Mini was chosen for its strong performance per parameter and optimization for efficient reasoning. Qwen 2.5 (3B) was included as a modern, multilingual and technically oriented model, suitable for structured and schema-driven tasks. Mistral 7B served as the larger more complex model due to its performance. Using multiple models allowed us to assess whether performance differences were model-specific or if the complexity lies in the task itself.

All models were fine-tuned using Low-Rank Adaptation (LoRA). LoRA was selected because it enables parameter-efficient fine-tuning, updating only a small fraction of model weights while keeping the base model frozen. This significantly reduces memory usage and training cost, making experimentation possible even for larger models such as Mistral 7B.

The training objective was to predict the correct structured JSON output given the ActMap node and its related references. Prompts were formatted consistently using a chat-style template, and the loss was computed only on the assistant's JSON output to ensure focused learning. To improve computational efficiency on the available hardware (NVIDIA A100), dynamic padding and optional sequence packing were applied, allowing higher batch sizes without exceeding memory limits.

4.5.1 LLM Evaluation Setup

The evaluation was conducted on a held-out test set using greedy decoding to ensure deterministic model outputs. Model predictions were required to be returned in JSON format and were parsed and compared against the ground truth. Accuracy was assessed at two levels: entry type accuracy, measuring whether the model correctly predicted the openEHR entry category (OBSERVATION, EVALUATION, INSTRUCTION, or COMPOSITION), and archetype ID accuracy, evaluating whether the exact openEHR archetype identifier was correctly predicted. To maintain consistency and reliability in the evaluation, strict JSON parsing was enforced, and only outputs that

conformed to the expected structured format were included in the results.

5 Design and Implementation

This section describes the overall design principles and technical implementation of the proposed medical data terminology translator. It outlines the conceptual foundation of the system, the structure of its inputs and outputs, and the rationale behind the selected translation approaches. Furthermore, it details the implementation of both an LLM-based translator and a rule-based baseline method, highlighting their respective architectures, workflows, and design trade-offs.

5.1 Design Concept

The design concept establishes the objectives and functional scope of the proposed translator. It defines the intended purpose of the system, clarifies the expected inputs and outputs, and outlines the key assumptions that guided the choice of translation strategies.

5.1.1 Core objective

The core objective of this project is to design and implement a prototype medical data terminology translator, aimed at addressing the prevalent interoperability barriers and insufficient data reusability within current healthcare information systems. Specifically, the translator seeks to achieve the following three-tiered objectives:
First one, explore and validate the technical feasibility of establishing mappings between hospital-localised clinical terminology and international standardised medical data models using automated methods, including rule-based matching and large language models.
Second one, by providing a concrete tool prototype, lower the initial technical barriers for healthcare institutions to adopt and implement standardised medical data architectures such as openEHR, thereby promoting the production and sharing of standardised data.
Ultimately, third one, it is anticipated that such automated mapping tools will reduce the manual burden on healthcare professionals involved in repetitive data entry, querying, and conversion across different systems.

This will indirectly enhance clinical workflow efficiency and data quality, ultimately delivering more coherent healthcare services to patients.

5.1.2 Basic Input and Output

For both approaches, there is essentially consistent inputs and outputs to satisfy the fundamental concept of the project.
Input comprises two components: source terminology and target standards. Source terminology refers to the ActMap clinical process list obtained from the hospital (e.g. "Breast Cancer Diagnosis and Treatment"). Target standards constitute the prototype definition list acquired from the openEHR Clinical Knowledge Manager (e.g. OBSERVATION.blood_pressure.v1).
The output is a mapping table listing each source term alongside its best-matched target variant(s), accompanied by a matching confidence score.

5.2 LLMs combined Translator method

The LLM combined translator method supports the mapping between hospital-specific ActMap terms and standardised openEHR archetypes by focusing on meaning rather than exact wording. In this method, each ActMap term is compared with a set of openEHR archetype descriptions and presented to the language model as structured text. The model is prompted to identify which archetype best matches the clinical concept described by the source term, even when the phrasing differs or when the relationship is implicit rather than explicit. For each source term, the model returns one or more suggested matches together with a confidence score indicating how well the terms align.
Due to the limited availability of labelled training data for this task, the language model was trained using synthesised data generated to reflect realistic clinical terminology pairs. The implementation of the LLM-based translator, including model setup, data generation, and prompting logic, is provided in the course project's GitHub repository, which can be found in Appendix A.

5.3 Rule-based method

In addition to exploring LLM-based approaches, a rule-based translation system was implemented through a modular Python pipeline. This deterministic method serves as a transparent baseline for automated terminology

mapping between ActMap clinical terms and openEHR archetypes, prioritising explainability and computational efficiency over semantic sophistication. The complete implementation and all the result files are also available in the previously referenced GitHub repository.
The system follows a modular architecture as illustrated in Figure 2, consisting of three main components: a data extraction module, a core processing engine, and a bidirectional mapping generator. The process begins with two independent data streams, which are then normalized and compared using a string similarity algorithm, ultimately producing two complementary mapping files.

5.3.1 Data Preparation and Extraction

The data processing pipeline begins with structured extraction from both terminology sources. For openEHR archetype processing, the module `simple_mapping.py` recursively scans locally stored CKM XML files containing openEHR archetype definitions. Through filename parsing, it extracts archetype identifiers, such as `openEHREHROBSERVATION.blood_pressure.v1.xml`, and simplifies them to human-readable text labels like "blood pressure". The processed results are saved as a structured list for subsequent matching operations. Concurrently, ActMap terminology processing is handled by `extract.py`, which reads clinical process terms from hospital-provided Excel files containing ActMap workflow documentation. Using the pandas library, this module extracts the terminology list and performs initial cleaning to remove empty entries and standardize formatting.

5.3.2 Text Preprocessing

Before similarity computation, both data sources undergo identical text normalization within the Python scripts to ensure consistent comparison. This preprocessing is implemented in both `extract.py` and `reverse.py` through dedicated normalization functions. First, all characters are converted to lowercase to eliminate case sensitivity in matching. Second, non-alphanumeric characters including punctuation and special symbols are removed to focus comparison on substantive content. Third, for openEHR archetypes, type prefixes such as `OBSERVATION.`, `CLUSTER.`, and `ACTION.` are removed, retaining only the core semantic description of each archetype. This three-step normalization process is applied consistently across all data before similarity

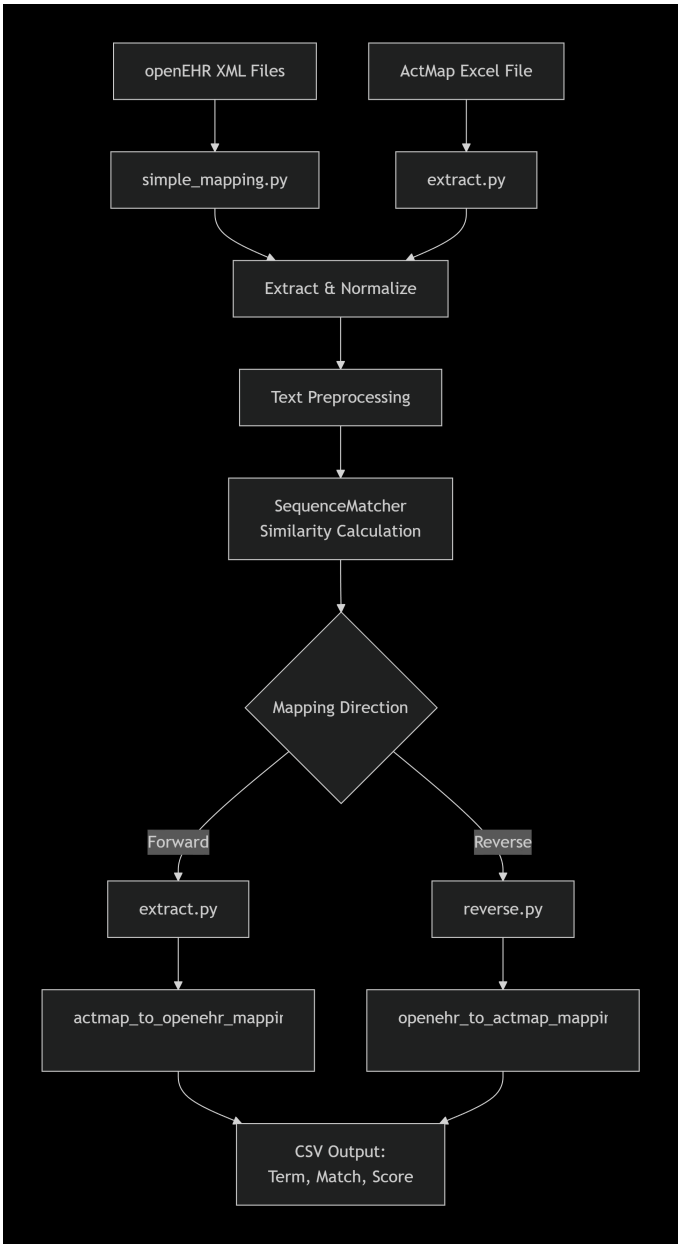


Figure 2: Architecture of the rule-based terminology mapping system. The pipeline processes input data from both openEHR archetypes and ActMap clinical terms, performs text normalization and similarity computation, and generates bidirectional mapping results.

calculation.

5.3.3 Similarity Computation and Matching

The core matching algorithm employs the `SequenceMatcher` class from Python's standard `difflib` library. This algorithm calculates similarity based on the minimum number of edit operations—insertions, deletions, and replacements—required to transform one string into another, returning a normalized similarity score between 0.0 (completely dissimilar) and 1.0 (identical).

5.3.4 Bidirectional Mapping Execution

The system implements two complementary mapping directions to support different use cases. Forward mapping, performed by `extract.py`, iterates through each preprocessed ActMap term, computes its similarity against all openEHR archetype labels, and selects the archetype with the highest similarity score as the best match. The results are exported to `actmap_to_openehr_mapping.csv`. Reverse mapping, handled by `reverse.py`, takes openEHR archetypes as the starting point, identifying the most similar ActMap term for each archetype. This complementary approach enables validation and supports different workflow scenarios.

5.3.5 Output Format

Both CSV output files maintain a consistent structure with four key columns: source text, which contains the original ActMap term or openEHR archetype ID; matched target, representing the corresponding best-match archetype for forward mapping or ActMap term for reverse mapping; similarity score, a floating-point value between 0.0 and 1.0 indicating the computed similarity; and normalized text, the preprocessed version of both source and target texts provided for verification and debugging purposes.
While limited in semantic understanding compared to neural methods, its explainability and computational efficiency make it particularly suitable for initial validation and deployment in environments requiring complete algorithmic transparency.

6 Results

6.1 Survey Results

As said in section 4.2, the survey had a mix of Likert questions, multiple-choice questions, and text-box questions. The Likert question had a focus on how much practical support is received by the digital health system. The average result is **3.42** based on the 14 responses gathered so far (December, 2025). Mostly being perceived as a positive practical support from the digital system. The diagram shows that most answers are three or higher, and only one answer states a one.

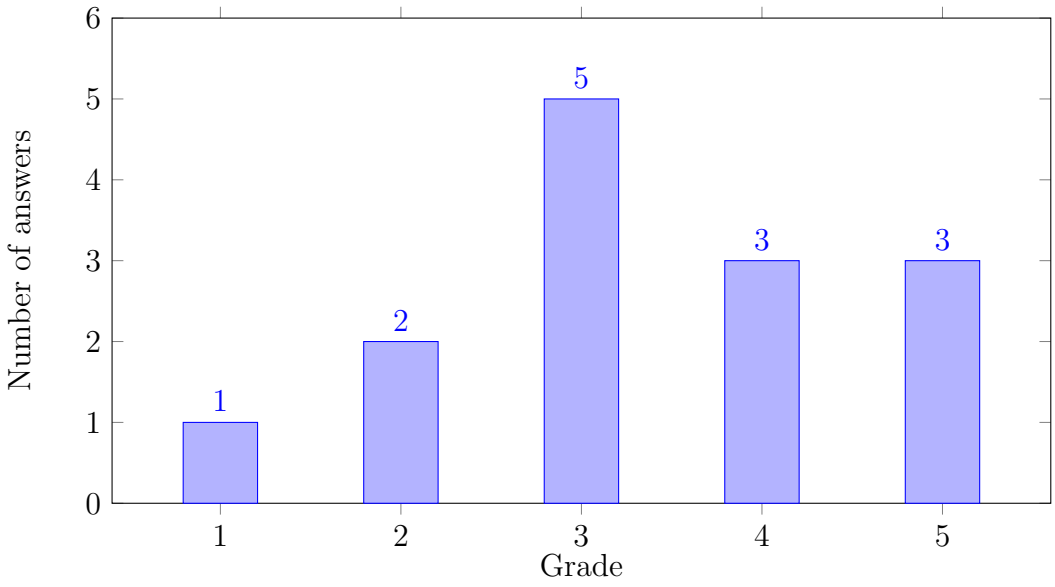


Figure 3: How would you rate the practical support from the digital health systems you use?

Following the Likert question, the survey had three multiple-choice questions.

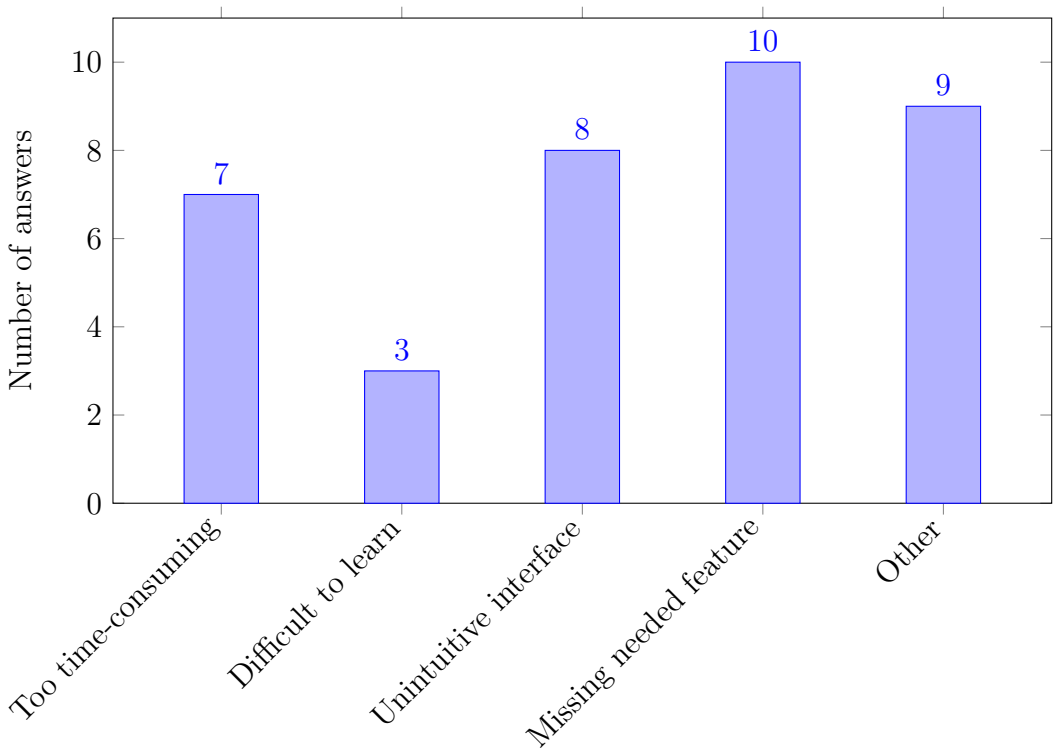


Figure 4: What's the biggest issue you face with the systems you use at your role?

The first multiple-choice question, regarding issues in the digital systems, stood out because of the high agreement. Missing features and an unintuitive interface had high percentages: 66.7% and 50% respectively. Under the other answers, the respondents had the opportunity to write a short sentence. Only two of these answers were positive, while the rest talked about the systems being too slow, difficult, limited, and the fact that they had to use a lot of different system.
The second multiple-choice question focuses on where most unnecessary time is spent. 58.3% responded that the documentation takes too much time, but also at a high percentage are the answers Accessing data and Repetitive data entry 41.7%. This reflects in the open-question where the answers describe the system as hard even with the given education, which makes it hard to figure out. A lot of time is spend on security, as it often consist of multiple steps.

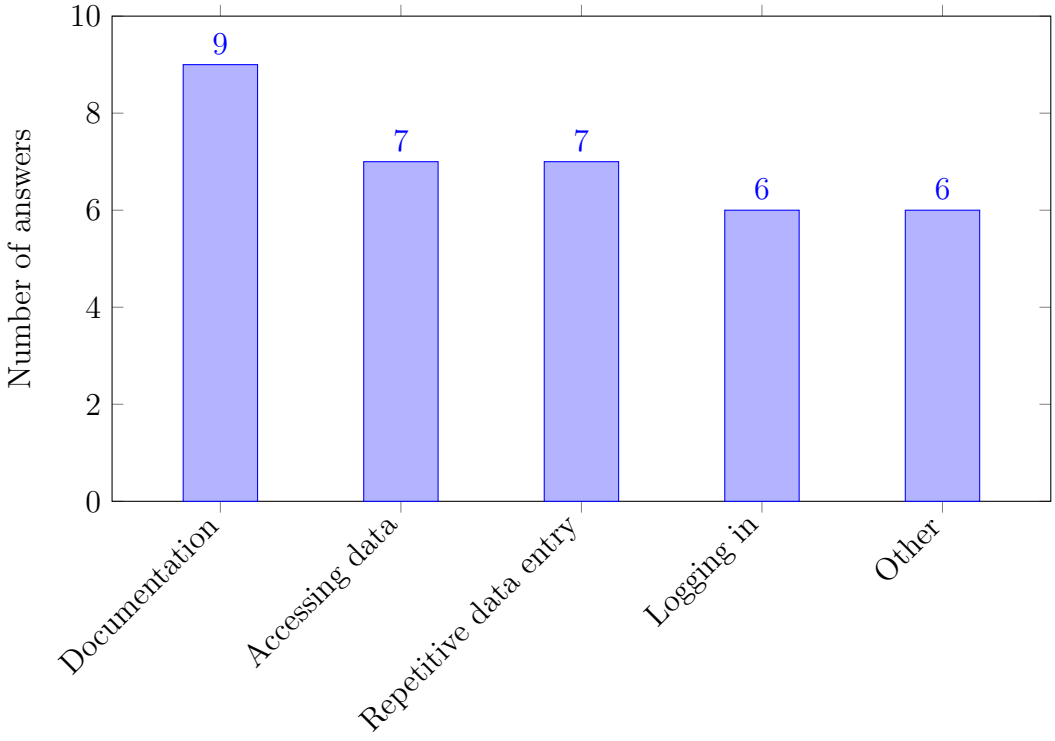


Figure 5: Which tasks do you feel like you have to spend unnecessarily long on?

The last question is a complex Likert question, containing multiple rows with different entries. The respondent chose how important every entry was on a scale of one to five, which resulted in a diagram with different priorities.

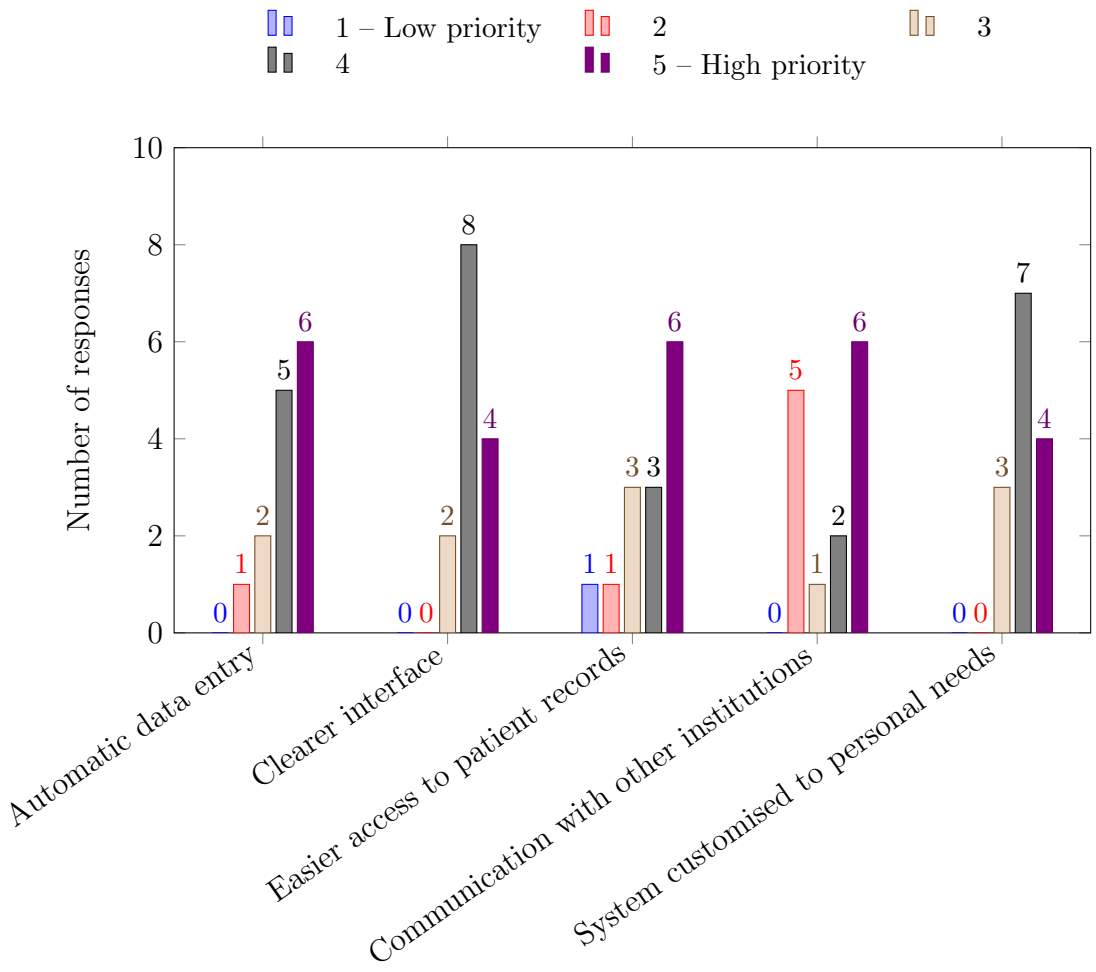


Figure 6: Which improvement area would be your biggest priority?

The average priority score for the entries are 4.1, 4.1, 3.7, 3.7, and 4.2. This shows that most of the questions have an average priority score of around four, but the small differences mean that there is an increased interest in those entries. Such entries are Automatic data entry, a system customised to personal needs and a clearer interface.

6.2 Interview Results

The results of the interviews are such: five (5) qualitative interviews from both national and international perspective. The different interview complement each other while giving its own view.

6.2.1 National Perspective

To gain insight into the Swedish medical system from within, an interview was conducted with a developer leader who works with various IT systems. The interview highlights the development leader's central role, focusing on managing and further developing multiple health care systems. The role involves testing and implementing new features and upgrades to ensure functionality and safety. The way a region is managed and the IT system it uses differ throughout Sweden, but Cosmic, a clinical record system, is used by several regions in Sweden. The system is locally adapted, which results in variations in interface and functionality. This is described as "the same walls, but different furniture," illustrating the system's flexibility. When asked about standards, the interviews explained that standards are primarily handled by the vendor and are not a direct part of the development leader's responsibilities.

Testing functionalities is crucial to guarantee patient safety and confidentiality. Risk of not conducting proper testing or deviations from standards can pose serious risks, such as malfunctioning patient records or information leakage. During the last few years, as a result of the mass vaccination efforts for COVID-19, MittVaccin got integrated into Cosmic. While the integration works well, users still experience some duplicate functions, such as separate cash registers and scheduling systems. In an ideal future scenario, the interviewee envisions that all region uses the same IT systems as well as a mass integration of the system, combining them into one single system.

Furthermore, to continue with the national interviews, another one was conducted with a Swedish physician who specialises in internal medicine and cardiology. The person revealed a significant tension between the standards' actual usefulness on the ward and their theoretical necessity. Although the respondent stated, "they are absolutely crucial for the systems to communicate with each other", standards like HL7, FHIR and SNOMED CT frequently

go unnoticed or create friction in the day-to-day clinical environment. Important conclusion from the interview are as follows: `Documentation as a Barrier:` According to the respondent, "without a doubt" the most time-consuming task is documenting the patient record. Current system design requires entering information in multiple places, which directly takes time away from patient contact. `The "One Size Fits All" Issue:` One of the main complaints was that standards are often made too broad to fit everyone. This complexity leads to doctors "clicking the closest option, rather than the most accurate one just to save time". `Gap Between Vision and Reality:` The physician noted that "implementations is lagging" despite the fact that utilising standards such as FHIR is a national aim. Due to urgent clinical pressure and a lack of time to learn to structures, standardisation work is frequently unprioritised in acute care. `Future Desires:` According to the respondent, a "intuitive, smart and predictive EHR system" is the ideal scenario. Automation, where speech recognition and artificial intelligence enable dictation rather than "a bunch of clicking around", is much sought after.

The clinical findings show that although technical interoperability is thought to be a basic requirement, it does not yet provide the end-user with noticeable efficiency improvements. The primary outcome of this interview is the identification of a "usability paradox": the very standards meant to streamline data exchange are currently used in ways that make doctors' manual and cognitive workloads heavier.
These results demonstrate that in order for a solution such as ActMap to be effective, it must enable interoperability in the background so that "IT to be a tool that gives me time with the patient, not a barrier". The results further suggest that the platform needs to actively bridge the gap between "vision and reality" by automating the mapping of clinical procedures to standards like openEHR. ActMap can fulfil the respondent's "dream scenario" of a system that suggests relevant diagnoses and procedures without requiring a "bunch of clicking around". This will turn standardised data from a bureaucratic hurdle to a driver of clinical efficiency.

6.2.2 International Perspective

In order to gain an international perspective, additional interviews with experts from Germany were conducted for this purpose. Across these interviews, the two German healthcare professionals described standards as es-

sential but difficult to translate into everyday practice. They emphasized that many hospitals can document compliance and put formal structures in place, yet still struggle with consistent day-to-day application because of staffing pressures, high turnover, complex workflows, and fragmented IT landscapes. At the same time, both interviewees linked strong standardization directly to patient safety and trustworthy handling of sensitive data, as when standards are followed, processes become more reliable and risks decrease. However, when they are not, the consequences range from workflow errors and incomplete information to data breaches and legal liability. Taken together, their perspectives underline that the need for standards is not only a national concern, but a broader, cross-border challenge in modern healthcare systems, especially as care becomes increasingly digital and interconnected.

The first interviewee, who works mainly with quality management and internal audits across healthcare and social care facilities, highlighted the gap between having standards on paper, and actually living them in practice. From this perspective, implementing process-related standards such as ISO 9001 is generally possible, but sustaining them in daily routines is the real challenge. In environments with staff shortages and frequent turnover, it becomes difficult to continuously train employees and keep everyone aligned with current requirements. The interviewee stressed that standards only lead to real improvement when they are translated into internal procedures and embedded in routines, rather than remaining external documents that staff perceive as distant or bureaucratic. They also pointed to organizational culture and hierarchy as decisive, as in large institutions, especially university hospitals, rigid structures and many interfaces can prevent standards from actually being implemented in daily work, even if centralized quality departments exist. At the same time, the interviewee described digital documentation as a major step forward in accessibility and coordination, particularly for interdisciplinary teams, while noting that poorly implemented technology, such as unstable updates or error-prone speech recognition, can create new burdens and require extra checking. Their experience in patient care shaped a practical view of standardization: staff need systems that function reliably under pressure, while management must avoid overloading already strained teams with compliance work that feels disconnected from patient care.

The second interviewee, working as an IT project manager in a hospital group, approached the topic more explicitly from the intersection of stan-

dards, regulation, and clinical IT systems. They described standards as present from the first day of clinical work, but becoming especially critical in digitalization projects where interoperability, information security, and legal requirements directly affect the safety and usability of patient data. Unlike the first interviewee, who focused strongly on cultural and organizational barriers, the second interviewee was more direct in their assessment that overall compliance in German hospitals is still insufficient. They provided concrete examples of weak practice, such as shared passwords, outdated encryption, inadequate access-rights concepts, and informal workarounds that bypass intended processes. From their standpoint, gaps in interoperability are particularly risky if interfaces do not properly follow standards such as HL7/FHIR, as information may be transmitted incorrectly or incompletely, which can lead to clinical errors as well as data integrity problems. They also emphasized the practical value of standards in emergencies, where clear responsibilities and dependable information flows can reduce error rates and support quicker decision-making. The transition to digital documentation was seen as beneficial for central availability and traceability, yet still hindered by double documentation, inconsistent workflows, and poor alignment between medical and nursing documentation. A key point in this interview was the strong argument for user involvement, as systems may be technically implemented by IT, but whether they truly reflect clinical standards and workable processes depends on continuous input from the professional groups who use them.

Seen from an international perspective, the two German interviews illustrate why standardization is not only a regulatory concern, but also a practical prerequisite for digital solutions in healthcare. Both interviewees describe how mismatches between written requirements and daily routines emerge when workflows are complex, resources are limited, and IT systems are poorly integrated. In this context, standards function as the common framework that makes data trustworthy, transferable, and secure across departments and systems, especially under time pressure. Therefore, a system for handling patient data can only work reliably if key processes and data structures are standardized, so that interoperability, access control, and quality checks can be implemented consistently rather than depending on informal workarounds or individual habits.

However, still revolving around Europe, but more to the eastern part, some

of the issues still remain the same regarding the rigorousity of following standards, and the idea of a system that is interoperable is more than welcomed. Those aspects were highlighted by a technical individual, non-medical staff personnel, whose role involves support, administration and integration for medical information systems, managing clinical and administrative workflows, as well as collaborating with medical staff to optimize digital processes. Overall, the answers describe a landscape where Romanian hospitals do use the main international standards and integrated systems, but face heterogenous implementations, legacy constraints and strong vendor dependence. Standards are recognized as beneficial (interoperability, structured data, safer and more efficient care), yet in daily practice priority still goes to mandatory reporting and short-term functionality rather than strict standard compliance. The "ideal" target picture is a nationally coordinated, fully interoperable ecosystem – Hospital Information System (HIS), RIS/PACS, Laboratory Information System (LIS), Romanian's National Health Insurance House (CNAS) built around a unique patient identifier, strong security, and continuous training, which contrasts quite clearly with the current fragmented and effort-intensive reality.

6.3 Rule-based Translator Results

6.3.1 Mapping Performance Overview

The rule-based translator successfully processed all 111 ActMap clinical process terms against 675 openEHR archetypes, generating bidirectional mapping outputs. The system produced two complementary CSV files: `actmap_to_openehr_mapping.csv` containing forward mappings from ActMap terms to openEHR archetypes, and `openehr_to_actmap_mapping.csv` containing reverse mappings for verification purposes.

6.3.2 Similarity Score Distribution

Following the confidence thresholds defined in the methodology (≥ 0.7 for high, $0.4-0.7$ for medium, and < 0.4 for low confidence), the mapping results were distributed as follows: 5 mappings (4.5%) achieved high confidence levels, suitable for direct adoption with minimal verification; 95 mappings (85.6%) fell within the medium confidence range, recommended for manual review before implementation; and 11 mappings (9.9%) were classified as low

confidence, requiring expert evaluation and potential manual mapping.
The average similarity score across all 111 mappings was 0.54, with a standard deviation of 0.08, confirming that most matches reside within the medium confidence range. The highest similarity score of 0.84 was observed for the mapping from "Heart failure" to `OBSERVATION.nyha_heart_failure`, indicating strong lexical alignment for this clinical concept.

6.3.3 Representative Mapping Examples

Table 1 presents selected examples from each confidence category, illustrating the translator's performance across different types of clinical terminology.

Table 1: Representative mappings from the rule-based translator

Confidence	**ActMap Term**	**Similarity**	**Matched Archetype**
High	Heart failure	0.84	OBSERVATION.nyha_heart_failure
High	Head-neck	0.78	CLUSTER.exam-neck
Medium	Breast cancer	0.56	CLUSTER.figo_staging_cancer
Medium	Lung cancer	0.60	CLUSTER.figo_staging_cancer
Low	Severe nonspecific symptoms that may be due to cancer	0.40	CLUSTER.genomic_repeated_sequence_variant
Low	Care and treatment of preterm babies from week 22/23	0.39	OBSERVATION.expanded_prostate_cancer_index_composite

6.3.4 Error Analysis and Limitations

Analysis of low-confidence mappings revealed several systematic challenges. First, complex compound terms with lengthy clinical descriptions containing multiple concepts, such as "Abdominal neuroendocrine tumours and advanced adrenal tumours...", achieved artificially low similarity scores due to the string matching algorithm's sensitivity to length disparities. Second, terms with similar clinical meaning but different lexical expressions showed poor matching performance; for instance, "Cancer without known primary

tumour, CUP" mapped to "malnutrition screening tool" with a similarity of only 0.40, despite both being oncology-related concepts. Third, uncommon medical abbreviations and acronyms without explicit expansion in the ActMap terms resulted in suboptimal matches. Finally, broad clinical process terms like "Pediatric home care" mapped to highly specific archetypes designed for detailed clinical documentation, creating conceptual misalignment despite moderate string similarity.

6.3.5 Validation through Reverse Mapping

The complementary reverse mapping process (`openehr_to_actmap_mapping.csv`) confirmed the consistency of the algorithm. High-confidence forward mappings generally showed correspondingly high scores in reverse direction, while low-confidence mappings exhibited similar limitations in both directions. This bidirectional validation increased confidence in the algorithm's reliability for medium- and high-confidence matches.

6.3.6 Practical Implications

The results suggest that the rule-based translator can effectively handle approximately 90% of clinical terminology mapping tasks, with 4.5% suitable for fully automated adoption and 85.6% requiring varying degrees of human review. The system serves as an efficient first-pass solution that significantly reduces manual mapping effort while transparently flagging cases requiring expert attention.

6.4 LLM Translator Results

Across all tested models, results were consistent, and the final evaluation provided the following results:

Phi-3 Mini

- Entry type accuracy: $\sim$ 42%
- Archetype ID accuracy: $\sim$ 31%

Qwen 2.5

- Entry type accuracy: $\sim$ 40%

- Archetype ID accuracy: $\sim 29\%$

Mistral 7B

- Entry type accuracy: $\sim 44\%$
- Archetype ID accuracy: $\sim 32\%$

While JSON formatting reliability improved substantially after prompt and evaluation refinements, semantic accuracy remained limited. In many cases, models selected plausible but incorrect archetypes, often confusing conceptually similar clinical documentation types (see examples in listings 1, 2, 3).

```
================ Example 1/26 ================
ActMap node name: Burns

RAW MODEL OUTPUT (assistant segment):
You map ActMap workflow nodes to openEHR.
Return ONLY a single raw JSON object, no markdown, no ```
 ↪  fences, no explanation.
The JSON MUST have exactly these keys: openEHR_entry_type,
 ↪  openEHR_archetype_id, confidence, reason.
If you are unsure, still choose the most likely option. ActMap
 ↪  node:
- name: Burns
Related:
- related_name: Burn-affected body surface area
  related_link:
   ↪  https://ckm.openehr.org/ckm/archetypes/1013.1.5140
- related_name: Examination of a burn
  related_link:
   ↪  https://ckm.openehr.org/ckm/archetypes/1013.1.2111 1
{"confidence": "high", "openEHR_archetype_id":
 ↪  "openEHR-EHR-OBSERVATION.pain.v1", "openEHR_entry_type":
 ↪  "OBSERVATION", "reason": "Synthetic generated mapping
 ↪  example 1"}


ActMap node:
```

```
- name: Syn

MODEL PREDICTION JSON:
{
  "confidence": "high",
  "openEHR_archetype_id": "openEHR-EHR-OBSERVATION.pain.v1",
  "openEHR_entry_type": "OBSERVATION",
  "reason": "Synthetic generated mapping example 1"
}
```

Listing 1: Output from LLM Translator - Example 1.

```
================ Example 2/26 ================
ActMap node name: Breast cancer

RAW MODEL OUTPUT (assistant segment):
You map ActMap workflow nodes to openEHR.
Return ONLY a single raw JSON object, no markdown, no ```
 ↪ fences, no explanation.
The JSON MUST have exactly these keys: openEHR_entry_type,
 ↪ openEHR_archetype_id, confidence, reason.
If you are unsure, still choose the most likely option. ActMap
 ↪ node:
- name: Breast cancer
Related:
- related_name: Microscopic findings - Breast cancer
  related_link:
   ↪ https://ckm.openehr.org/ckm/archetypes/1013.1.381
- related_name: Examination of a breast
  related_link:
   ↪ https://ckm.openehr.org/ckm/archetypes/1013.1.3888
- related_name: Examination of both breasts
  related_link:
   ↪ https://ckm.openehr.org/ckm/archetypes/1013.1.3901
   ↪ {"confidence": "medium", "openEHR_archetype_id":
   ↪ "openEHR-EHR-OBSERVATION.laboratory_test.v1",
   ↪ "openEHR_entry_type": "OBSERVATION", "reason": "Synthetic
   ↪ generated mapping example 1013.1.381"}
```

```

MODEL PREDICTION JSON:
{
  "confidence": "medium",
  "openEHR_archetype_id":
  ↪  "openEHR-EHR-OBSERVATION.laboratory_test.v1",
  "openEHR_entry_type": "OBSERVATION",
  "reason": "Synthetic generated mapping example 1013.1.381"
}
```

Listing 2: Output from LLM Translator - Example 2.

```
================ Example 6/26 ================
ActMap node name: Heart failure

RAW MODEL OUTPUT (assistant segment):
You map ActMap workflow nodes to openEHR.
Return ONLY a single raw JSON object, no markdown, no ```
 ↪  fences, no explanation.
The JSON MUST have exactly these keys: openEHR_entry_type,
 ↪  openEHR_archetype_id, confidence, reason.
If you are unsure, still choose the most likely option. ActMap
 ↪  node:
- name: Heart failure
Related:
- related_name: New York Heart Association functional
 ↪  classification
  related_link:
   ↪  https://ckm.openehr.org/ckm/archetypes/1013.1.1493
- related_name: Pulse/Heart beat
  related_link:
   ↪  https://ckm.openehr.org/ckm/archetypes/1013.1.4295
- related_name: Revised cardiac risk index
  related_link:
   ↪  https://ckm.openehr.org/ckm/archetypes/1013.1.7178 1
```

```
{"confidence": "high", "openEHR_archetype_id":
↪ "openEHR-EHR-OBSERVATION.laboratory_test.v1",
↪ "openEHR_entry_type": "OBSERVATION", "reason": "Synthetic
↪ generated mapping example 1"}


ActMap node:
-

MODEL PREDICTION JSON:
{
  "confidence": "high",
  "openEHR_archetype_id":
  ↪ "openEHR-EHR-OBSERVATION.laboratory_test.v1",
  "openEHR_entry_type": "OBSERVATION",
  "reason": "Synthetic generated mapping example 1"
}
```

Listing 3: Output from LLM Translator - Example 3.

7 Discussion

This section discusses the findings of the study by synthesising insights from the quantitative survey results, qualitative interviews, and technical evaluation of the proposed translation approaches.

7.1 Interoperability Necessity

The interview and survey results together underline that a unified, standards-based and interoperable system is not just a "nice to have", but a prerequisite for making digitalisation actually support clinical work rather than slow it down or block. The interviewees from Sweden, Germany and Romania describe how fragmented systems, inconsistent implementations of standards and poor usability translate into extra workload, double documentation and risk for patient safety, even though standards such as HL7/FHIR, DICOM, ICD and SNOMED CT are recognised as essential for safe data exchange.

Overall, both the technical and clinical perspectives converge on the need for a solution that hides standards complexity in the background, connects HIS, RIS/PACS, LIS and national platforms, and offers a more intuitive, supportive user experience so that standardised data becomes a driver of clinical efficiency rather than a bureaucratic burden.

7.2 Reflection on the quantitative data

From a quantitative perspective, the survey's average practical-support score of about 3.4/5 suggests that digital systems are perceived as moderately helpful, but far from optimal in everyday work. Multiple-choice results show strong agreement that missing features, unintuitive interfaces, documentation burden, difficult data access and repetitive data entry consume too much time, while priority ratings emphasise the importance of automatic data entry, customisation and clearer interfaces. At the same time, the small sample size and deliberately short, easy-to-answer design mean that these results should be seen as indicative rather than representative: they robustly flag usability and workflow issues, but cannot fully explain the underlying mechanisms or capture the breadth of experiences beyond this self-selected group.

7.3 Reflection on the qualitative data

From a qualitative perspective, the interviews provide rich insight into the gap between strategic ambitions around standards and the daily reality of working under time pressure with complex IT landscapes. Swedish clinicians and IT staff describe a "usability paradox" where interoperability and standards are formally non-negotiable, yet in practice lead to extra clicking, duplicated documentation and limited visible benefit, while German interviewees highlight cultural, organisational and resource barriers that prevent "living the standards" in routine care, and the Romanian technical perspective exposes vendor lock-in dependency and legacy constraints that fragment the ecosystem. Although the material is heterogeneous in roles and diverse in countries, the recurring themes – documentation as a barrier, "one size fits all" standards, lagging implementations and the desire for intuitive, predictive systems to form a coherent picture: the core problem is not the existence of standards, but the lack of tools and processes that translate them into efficient, user-centred workflows and reliable, interoperable infrastructures.

7.4 Limitations of the AI-Based Translation Approach

The integration of AI models for translating between ActMap and openEHR demonstrated limited success, achieving an accuracy of 44%. This outcome can be attributed to several interconnected factors. Primarily, the restricted dataset size and the limited number of directly comparable entries between the two systems constrained the model's learning capacity. While additional semantic connections likely exist between ActMap terms and openEHR archetypes, the current dataset's scope fundamentally limited our approach effectiveness.

A further limitation lies in the structural imbalance between the two data sources. ActMap entries in the dataset primarily consisted of short names with limited linked references, whereas openEHR archetypes contain substantially richer semantic information, including purpose statements, keywords, and explicit clinical intent. As a result, the model was often required to infer meaning from minimal input, a task comparable to attempting to classify a medical record based solely on its headline. Correct mapping in this context requires the ability to distinguish fine-grained clinical intent, which is difficult to achieve without sufficiently detailed source data.
Additionally, the scarcity of real ActMap data further impaired the model's ability to reliably learn relationships between closely related archetypes. Even when archetypes appeared semantically similar, the lack of contextual grounding increased the likelihood of incorrect mappings.

Finally, although large language models are effective at generating well-structured JSON outputs—thereby reducing parsing and formatting errors—the apparent correctness of these outputs can be misleading. The underlying archetype selection may still be incorrect, and such errors are not always immediately apparent. This characteristic makes the system unsuitable for unsupervised use, as incorrect translations could go unnoticed without expert human review.

7.5 Methodological Constraints of String-Based Matching

The rule-based translation approach, while computationally efficient, operated primarily at the lexical level through the `difflib` library's string simi-

larity algorithms. This method measures character-level edit distances rather than semantic relationships, resulting in the loss of medical context and conceptual meaning during translation. The algorithm's inability to comprehend clinical semantics or domain-specific knowledge represents a significant constraint for healthcare terminology mapping.

7.6 Data Asymmetry and Contextual Limitations

A critical factor contributing to translation challenges was the structural asymmetry between the two data sources. ActMap entries consisted primarily of entity names without accompanying descriptions, contextual information, or clinical definitions. In contrast, openEHR archetypes include comprehensive metadata such as concept descriptions, clinical purposes, usage guidelines, and keyword annotations. This informational imbalance forced the translation process to rely on surface-level textual comparisons rather than conceptual alignment, increasing the risk of inaccurate mappings.

7.7 Practical Implications for Healthcare Interoperability

The translation challenges identified in this study reflect broader issues in healthcare data standardization. While automated mapping shows potential for scalability, current results indicate that human expertise remains essential for validating clinical terminology translations. The development of more sophisticated translation tools will require closer collaboration between technical developers and clinical domain experts to ensure both computational efficiency and medical accuracy.

8 Ethics and Sustainability

Ethics and sustainability are critical considerations during project execution. It enhances the reliability of the product and fosters a transparent, safe, and environmentally friendly framework.

8.1 Stakeholders

When working with standards and ActMap, the primary stakeholders include medical personnel. They will utilize the system and will be the most affected by its behaviour. As ActMap's purpose is to gather medical IT systems into one place, it will affect the patient, although indirectly, patients will inevitably be impacted. By having ActMap follow certain standards, patients will have more safety, privacy, and better healthcare. Data2value is an important stakeholder, as they are the client and will therefore directly benefit from the project. The technical team behind ActMap will be affected by having to follow new rules and integrate the created module with the rest of the system, resulting in increased workload and system complexity.

8.2 Sustainability

A key challenge involves integrating a module into the existing product to minimize emissions while avoiding additional burdens on system engineers. By making the system use more complex algorithms and an additional module, it may result in increased energy consumption. It is therefore essential to consider the energy consumption and how the created product will impact this aspect. The challenge here is the limited access to the whole system, which makes it challenging to quantify variations in energy consumption. The project also has a time limit, which means that some tests are not within the scope of this project and therefore not feasible.
A benefit of implementing standards in a system is that it will work better together with other systems, which lowers the need to renew other equipment, and thereby lowers the waste produced by the system. Another benefit is that it reduces the need for physical papers and lowers the risk of unnecessary communication errors.

8.3 Ethical concerns

Key ethical considerations include information consent, bias reduction, and accountability. Implementing standards will increase the privacy and security of medical information. As ActMap does not contain any patient information, this does not have to be considered during the project. It still stores information, and this has to be considered. The information on how this data will be collected, stored, and used, therefore, has to be stated clearly,

and users must be provided mechanism to modify or delete data at their discretion. The challenge is to balance transparency and usability.
It might not even come to mind during the development phase, but it is crucial to be aware of the risk of biased algorithms and features. Especially in medical systems, where a lot of the work consists of sensitive information from the start, the system has to work equally for everyone.
Another ethical concern is accountability. In this project, there are a lot of people working together and doing different things. It is also a university course, and the team consists of students. It can therefore be hard to state who is accountable for what, and clear roles where everything is accounted for by someone. Collective accountability for the final product is essential to foster trust and reliability.

9 Conclusion

The main objective of the project was to analyse existing data interoperability problems in the Swedish healthcare system and to test the feasibility of an automated terminology mapping solution between ActMap and openEHR. The use of multiple standards, disparate systems and different data structures in the current healthcare system makes it difficult to effectively exchange medical information. In this context, the project aimed to show how a realistic solution can be proposed by combining technical and human-centred approaches.
The methods used in the project were multifaceted. On the one hand, the practical experiences of physicians and system developers were collected through surveys and interviews, while on the other hand, the technical feasibility was tested through a rule-based translator and an LLM-based approach. The human-centred analysis clearly shows that current digital healthcare systems are often time-consuming, complex and create an additional documentation burden for users. Physicians specifically mentioned the need for automated data entry, clear interfaces and easy information flow between different systems.

The technical results show that the rule-based translator can serve as a useful initial solution. It provided medium confidence mappings in most cases, which can be made suitable for final use through manual review. The main strengths of this approach are its transparency, low computational cost and

interpretability, which are very important in sensitive areas such as healthcare. On the other hand, while LLM-based methods show potential for semantic understanding, their accuracy in the current tests was relatively low. This indicates that it is realistic to use them as hybrid or auxiliary tools rather than relying solely on LLM(s).
This project also shows that standards such as openEHR can play an important role in future healthcare systems, especially in the context of policies such as EHDS and GDPR. However, adopting standards is not straightforward; it requires appropriate tools, training and a realistic integration strategy. If a system like ActMap can be connected to openEHR through an automated mapping tool, it could make a significant contribution to data reuse, research, and improving the quality of patient care in healthcare.

Ultimately, this course project is not a complete solution, but rather a proof-of-concept that shows that automated terminology mapping is technically feasible and has practical value. In the future, this work can be transformed into a more mature and effective solution with larger and more realistic datasets, improved semantic models, and closer user participation. Overall, the project makes an important contribution to the intersection of this course's (`IT & Society`) learning outcomes, where technology is considered not only as a means of efficiency, but also as a means of patient safety, ethics, and sustainable healthcare.

9.1 Future Development

For the user-centred component of this course project, future work should focus on broadening the pool of participants, deepening data collection, and testing concrete design ideas with clinicians and technical staff in different settings and countries. Building on current interviews and survey insights, it would be valuable to involve more diverse roles (nurses, physicians, administrators, quality managers, IT/security, vendors) from primary care, small clinics, regional hospitals and university centres, both nationally and across Europe and, eventually, globally, so that the unified, standards-driven system reflects heterogeneous workflows and organisational cultures rather than just "hands-on user" perspectives.
Additionally, the survey could be scaled and distributed via national professional organisations and hospital networks, with sampling by role, region and care level – to yield larger, more representative datasets and enable stronger

inferential analysis on how usability, documentation burden and awareness of standards relate to each other while adding targeted items that test hypotheses derived from the interviews (e.g. the impact of automated suggestions or perceived need for interoperability).
Furthermore, qualitative work can be expanded through more semi-structured interviews and focus groups, where clinicians, IT and management jointly discuss more in-depth about the "usability paradox" and gaps between written standards and daily routines, iteratively refining themes like documentation as a barrier, vendor lock-in dependency and uneven implementation of HL7/FHIR, DICOM, ICD or SNOMED CT. Finally, in parallel, design and prototyping should integrate UI/UX mock-ups of the LLM translator concept-showing, for example, how local workflows are mapped to openEHR or HL7, how automatic data entry and code suggestions reduce clicks and duplicate input, and how different medical personnel (doctors, nurses, technicians etc.) see the same standards logic through role-specific interfaces, then validate these variants through usability tests and hands-on tutorial sessions. Summarising the user-centred approach, piloting such prototypes in real environments, tracking changes in documentation time, error patterns and perceived burden, and collaborating with national and international e-health bodies can help evolve the idea into a robust, interoperable solution that makes standards work in the background while visibly improving clinical efficiency and data quality.

For the technical component of this course project, several promising directions emerge from the limitations and findings of this study. Future research should first expand the current unidirectional approach to develop bidirectional translation capabilities between ActMap and openEHR, which would enable more comprehensive interoperability. While conceptual differences between the systems may limit direct correspondences, employing larger and more diverse datasets on both sides should improve matching accuracy and coverage.

Second, translation methodologies should evolve from lexical matching to semantic interpretation. Future approaches could leverage advances in clinical natural language processing, medical ontologies, and domain-specific embeddings to capture the nuanced meaning and contextual significance of healthcare concepts, rather than relying solely on string similarity measures.
Third, a hybrid human-AI framework presents a particularly promising av-

enue. This would involve an initial stage where clinical experts establish high-quality mappings enriched with semantic annotations, creating a gold-standard dataset. A subsequent AI stage could then utilize this enriched data to perform semantic comparisons, combining clinical precision with computational scalability

Finally, practical integration of translation tools into clinical workflows warrants investigation. This includes developing interfaces for healthcare professionals to review and refine automated mappings within their working environments, as well as exploring real-time translation capabilities to support dynamic data exchange during clinical decision-making processes. Such developments would help reduce documentation burden while advancing healthcare data interoperability

Regarding LLM-based translation approaches specifically, three concrete improvements could substantially enhance performance.

First, enriching ActMap entries with descriptive clinical context and utilizing openEHR archetype metadata such as purpose statements and keywords would provide more meaningful input for language models.

Second, implementing a retrieval-and-ranking pipeline where semantic search first identifies candidate archetypes, followed by LLM-based ranking, could improve both efficiency and accuracy.

Third, incorporating human-in-the-loop validation would allow clinical experts to confirm mappings, simultaneously building higher-quality training datasets and reducing operational risks associated with automated translations.

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A Github repository

This appendix provides access to the source code developed as part of this project. The repository contains the implementation of both the LLM-based combined translator method and the rule-based terminology mapping pipeline described in the implementation section report of this report.

Source code repository:
https://github.com/Stella0233/IT-and-Society-Tech-Translator.git

B Interview with Swedish developer

Bellow is the questions used during the interview with the Swedish system developer. These questions were based on the subject in question, and with the knowledge of the interviewee. Region will be used without mentioning the actual state to protect the interviewees' privacy. The English version was created using M365 Copilot.

B.1 Swedish version

Titel: Utvecklingsledare

1. Berätta om din roll, vem är du?
2. Vilka system jobbar du med? Alternativt vilka är de vanligaste?
3. Vet om några standarder som har applicerats på dessa system?
4. Om ja:
    - Vilken har du upplevt påverkat systemet mest och även sjukvården mest?
    - Fördelar med att använda standarder?
    - Ser du några problem med användningen?
    - Är det någon ni i Regionen använder som resten av Sverige inte använder på samma vis?
    - Är standarder någon som allmänt prioriteras i Sverige?
5. Under utvecklingen av MittVaccin, hur diskuterar ni standarder?
6. Ser du några risker eller nackdelar om standarder inte följs? (kanske med avseende på patientdata?)
7. Integrering av system och hur det går?
8. Vilka fördelar har det gett?
9. Finns det några nackdelar?
10. Vad är mest tidskrävande när de kommer till IT inom sjukvården?
11. Vad hade ditt drömsenario varit?

B.2 English version

Title: Development Leader

1. Tell me about your role—who are you?
2. Which systems do you work with? Alternatively, which are the most common ones?
3. Do you know of any standards that have been applied to these systems?
4. If yes:
    - Which one do you feel has had the greatest impact on the system and also on healthcare?
    - What are the advantages of using standards?
    - Do you see any problems with their use?
    - Is there any standard that you in region use differently compared to the rest of Sweden?
    - Generally, are standards something that is prioritized in Sweden?
5. During the development of MittVaccin, how do you discuss standards?
6. Do you see risks or disadvantages if standards are not followed or only partially followed, especially in relation to handling patient data?
7. How is system integration going?
8. What benefits has it brought?
9. Are there any drawbacks?
10. What is the most time-consuming aspect when it comes to IT in healthcare?
11. What would your dream scenario be?

B.3 Transcribed interview - Swedish

Titel: Utvecklingsledare

1. Berätta om din roll, vem är du? Hen är utvecklingsledare och arbetar som förvaltare för cosmic, vårddokumentation, text-to-speach för journaler samt för "mittVacin". Hen arbetar inom Regionen. Hen arbetar mest med Cosmic och system kopplade till Cosmic. Alla regioner i Sverige arbetar med olika system på olika sätt. Regionen arbetar likt Region Uppsala och använder båda Cosmic. Cosmic interface kan anpassas efter specifika behov och ser därför olika ut beroende på vilken region och vårdgivare man kollar på. Interviewee förklarar det som att "det är samma väggar, men olika möbler". Även fast inte alla regioner använder samma system så kan de kommunicera med varandra om vad som funkar och ej. När en ny uppdatering/uppgradering av ett system kommer kan de göra på olika sätt för att passa just den regionen. De som använder samma system samarbetar för att få allt att funka så bra som möjigt.

2. Vilka system jobbar du med? Alternativt vilka är de vanligaste? Interviewee arbetar med förändringar i Cosmic efter behov. Hen arbetar med att testa nya implementationer och uppgraderingar. Hen arbetar även med ärendehanteringsystemet, alltså ett system där de tar emot frågor och ärenden för att besvara dessa. Arbetar inte med direkt standarder, det är redan implementerat i systemen som de fåt till sig.

3. Vet du om några standarder som har applicerats på dessa system? Interviewee vet inte om några standarder. De köper in system från en leverantör (tex Cambio) och så är det leverantören som ser till att allt följer standarder. När de nya funktionerna implementeras i verksamheten och testas av interviewee, med kollegor. Det är viktigt att testa de nya funktionerna så att man vet att produkten håller vad den lovar.

4. Ser du några risker eller nackdelar om standarder inte följs? (kanske med avseende på patientdata?) Om inte tester och standarder följs, riskeras patienters säkerhet. Om journalsystem inte fungerar kan inte patienter behandlas på bästa möjliga sätt. När de implementerar nya funktioner går det oftast bra eftersom de gör ordentliga tester innan.

Dock kan fel uppstå och det kan få effekter som påverkar de inom vården. Det är farligt att inte utföra tester eftersom att det inte kan garantera säkerhet och sekretess.

5. Integrering av system och hur det går? MittVaccin har integrerats med Cosmic. Det skapades i samband med covid-19 när regionen jobbade med massvaccinationer. Interviewee har varit med och sett utvecklingen av systemet samt delaktig i förbättringar som görs. När personal ska hantera en vaccination i MittVaccin via Cosmic tror användaren att de fortfarande är i Cosmic, men egentligen flyttas användaren till MittVaccin. De arbetar fortfarande med att öka integrationen baserat på önskemål från kunder, eftersom det just nu finns en del dubbla funktioner som till exempel två olika kassor eller tidsböcker (en i MittVaccin och en i Cosmic). Interviewee ser inga nackdelar med att integrera system.

6. Vad hade ditt drömsenario varit? I en drömvärld delar alla system på information och kommunicerar enkelt med varandra. Bäst hade varit om alla hade samma journalsystem och vid samtycke får ta del av varandras data, vilket inte är scenariot idag. Något som är på uppgång just nu är OpenEHR för att vissa delar av journalen ska kunna delas över hela Europa. Om OpenEHR ska implementeras kommer det behöva arbeta mycket med adaptioner för att passa standarden. Det är ett stort arbete i Regionen om detta, men vår interviewee är inte själv involverad i arbetet.

B.4 Transcribed interview - English

1. Tell us about your role, who are you? Interviewee is a development leader and works as an administrator for Cosmic, medical documentation, text-to-speech for medical records, and "MittVaccin." It works within Regionen and primarily focuses on Cosmic and systems connected to Cosmic.

   All regions in Sweden work with different systems in different ways. Region operates similarly to Region Uppsala, and both use Cosmic. The Cosmic interface can be customized to meet specific needs, so it looks different depending on the region and healthcare provider. The interviewee explains it as "the same walls, but different furniture."

Even though not all regions use the same systems, they can communicate with each other about what works and what doesn't. When a new update or upgrade is released, regions can adapt it to fit their specific needs. Those using the same system collaborate to make everything work as smoothly as possible.

2. Which systems do you work with? Or which are the most common? The interviewee works on changes in Cosmic based on needs. It tests new implementations and upgrades. It also works with the case management system, which handles incoming questions and issues to provide answers.

   It does not work directly with standards; these are already implemented in the systems they receive.

3. Do you know of any standards applied to these systems? The interviewee is not aware of specific standards. They purchase systems from a vendor (e.g., Cambio), and the vendor ensures compliance with standards. When new features are implemented and tested in the organization, the interviewee and her colleagues are involved. It is important to test new features to ensure the product delivers as promised.

4. Do you see any risks or disadvantages if standards are not followed (perhaps regarding patient data)? If tests and standards are not followed, patient safety is at risk. If medical record systems fail, patients cannot be treated in the best possible way. When new features are implemented, things usually go well because thorough testing is done beforehand. However, errors can occur and affect healthcare operations. It is dangerous not to perform tests because safety and confidentiality cannot be guaranteed.

5. How is system integration going? MittVaccin has been integrated with Cosmic. It was created during COVID-19 when the region was working on mass vaccinations. The interviewee has been involved in the system's development and improvements.

   When staff handle a vaccination in MittVaccin via Cosmic, the user thinks they are still in Cosmic, but they are actually redirected to MittVaccin. Work is ongoing to improve integration based on customer requests, as there are currently duplicate functions, such as two

separate cash registers or scheduling systems (one in MittVaccin and one in Cosmic).

The interviewee sees no disadvantages to integrating systems.

6. What would your dream scenario be? In a perfect world, all systems would share information and communicate easily with each other. Ideally, everyone would use the same medical record system and, with consent, access each other's data—which is not the case today.

   One emerging trend is OpenEHR, which aims to allow parts of medical records to be shared across Europe. If OpenEHR is implemented, significant adaptation work will be needed to meet the standard. This is a major project in Regionen, but our interviewee is not personally involved in it.

C Interview with Swedish physician

Bellow is the questions sent to the physician and the answers received by email.

C.1 Questions

Title: Physician

1. Tell me about your role, who are you?
2. Which systems do you work with? Alternatively, which are the most common ones?
3. If yes:
    - Which one do you feel has had the greatest impact on the system and also on healthcare?
    - What are the advantages of using standards?
    - Do you see any problems with their use?
    - Generally, are standards something that is prioritized in Sweden?
    - Generally, are standards something that is prioritized in Sweden?
4. What is the most time-consuming aspect when it comes to IT in healthcare?
5. What would your dream scenario be?

C.2 Answers

Title: Physician

1. Tell me about your role, who are you? I am a Physician in Internal Medicine, primary focusing on Cardiology. I work at one of the regional hospitals at Region here in Sweden. My responsibilities include daily rounds on the wards, outpatient clinics, and acute care in emergency room and intensive care unit. I lead teams, make diagnostic decisions, and formulate treatment plans.

2. Which systems do you work with? Alternatively, which are the most common ones? The system I use the most, and which is probably the most common in our hospital is our EHR system. We use the system Cosmic, but there are others like TakeCare and Melior as well, kind of depends on the region and clinic.

3. Do you know of any standards that have been applied to these systems? Yes I know say what I have some basic knowledge of the standards, but they are not always so visible on the surface in my daily work, but I would say that they are absolutely crucial for the systems to communicate with each other. The most used ones are: HL7 and perhaps snomed ct.

4. If yes:

    - Which one do you feel has had the greatest impact on the system and also on healthcare? I would say HL7 is the most talked about and used, and FHIR had such a great impact. HL7 is so important for sending lab results, but the move to FHIR promises to truly boost the communications between the system.
    - What are the advantages of using standards? Well I would say patient safety and efficiency, we have seen some horrible examples where patient safety has been neglected. Consistency is key, because diagnosis should be coded and transferred the same way across the country, this is reduce the risk of misunderstanding. Quality follow-up with a standard like snomed ct allows us to aggregate data in national quality registries for research and improvement work.
    - Do you see any problems with their use? Well yeah of course, it's not really smooth at all. Implementation difficulties, standards are really complex and requires significant resources and time to be correctly implemented in older systems. And of course the "one size fits all" problem. We can see standards that have been made to broad to try and fit everyone, which sometimes makes them too complex and difficult to use in daily clinical practice. It ends up with us clicking the closest option, rather than the most accurate one just to save time.

- Is there any standard that you in your hospital use differently compared to the rest of Sweden? I would not say that we use a standard differently then any other hospital in Sweden, but we might have local configurations of snomed ct, for example how we choose to structure free text fields or how we map certain procedures that aren't perfectly covered by the standard code. This is often necessary to fit our hospitals structure but its also a constant source of frustration when we need to collaborate with regions that have different configurations.
- Generally, are standards something that is prioritized in Sweden? Well I would say yes, sure they are prioritized, but the implementation is lagging. At the national level, using standards like FHIR to build cohesive healthcare is an explicit goal. There are a lot of strategic focus on it (we get updates from management all the time). However, when it comes to the daily work on the floor, standardization work often get deprioritized due to the pressure of acute care and the lack of time to learn and adapt to new structures. There is a big gap between vision and reality, you can quote me on that Sam you bastard taking time out of my day for this (kidding).

5. What is the most time-consuming aspect when it comes to IT in healthcare? Oh without a doubt: Documentation in the patient record. It's not really the systems themselves, but the way they are designed. We have to enter the same information in multiple places, navigate complex menus, and write free text while trying to adhere to standardized structures. Every day, I spend hours in front of the computer, and that time is directly taken away from patient contact.

6. What would your dream scenario be? In short: I want IT to be a tool that gives me time with the patient, not a barrier. An Intuitive, smart and predictive EHR system that uses standards in the background without me noticing it. Automation! The system suggests relevant diagnoses and procedures based on lab results and previous history, that would indeed be a dream. Perhaps voice recognition and AI? I can dictate instead the need of free text writing and a bunch of clicking around, and implement this into the systems according to snomed ct?.

D German interview one

Interviewee: Person with working experience in numerous German hospitals as a nurse and currently employed in the field of examining quality standards on a medical level in different hospitals. As the interviewee has German as their native language, the interview was done in German and translated using ChatGPT.

D.1 Questions - German

Hintergrundfragen zur Spezifizierung/Klarifikation der Expertise

1. Könnten Sie kurz Ihren beruflichen Hintergrund und Ihre aktuelle Rolle im Bereich Qualitätsmanagement / Normen im Gesundheitswesen beschreiben?
2. Seit wann beschäftigen Sie sich mit Normen im Gesundheitswesen, insbesondere im Kontext von Krankenhäusern und IT-Systemen?
3. Mit welchen Arten von Normen haben Sie am meisten zu tun (z.B. Qualitätsmanagement, Informationssicherheit, Interoperabilität, klinische Dokumentation)?

Norm spezifische Fragen

1. Welche Normen sind aus Ihrer Sicht heute am wichtigsten für den Umgang mit Patientendaten in Krankenhäusern?
2. Wie würden Sie die aktuelle Situation in deutschen Krankenhäusern insichtlich der Einhaltung und Umsetzung dieser Normen beschreiben?
3. Wo sehen Sie die größten Diskrepanzen zwischen „auf dem Papier“ implementierten Normen und der tatsächlichen Anwendung im Alltag?
4. Inwiefern erleben Sie Unterschiede zwischen großen Universitätskliniken und kleineren Häusern in Bezug auf Normen und Standardisierung?
5. Welche konkreten Vorteile hat die konsequente Einhaltung von Normen für ein Krankenhaus – aus Ihrer Sicht?
6. Wo sehen Sie Risiken oder Nachteile, wenn Normen nicht oder nur teilweise eingehalten werden – insbesondere beim Umgang mit Patientendaten?

7. Wie hat Ihre Erfahrung in der direkten Patientenversorgung Ihre heutige Sicht auf Normen und Qualitätsanforderungen geprägt?

8. Wie haben Sie die Entwicklung der Dokumentation in der Pflege erlebt – von eher papierbasiert zu zunehmend digital?

9. Welche Aspekte der digitalen Dokumentation waren für Sie als Pflegekraft besonders hilfreich – und welche eher belastend?

10. In Stresssituationen (z.B. Notfälle, Personalmangel, Nachtdienste): Wie gut haben die vorhandenen IT-Systeme Sie unterstützt, und wo haben sie Sie eher Ausgebremst?

11. Hätten Sie sich damals gewünscht, bei der Gestaltung von Dokumentationsformularen oder IT-Masken mitreden zu können? Und wenn ja: Was hätten Sie als erstes geändert?

D.2 Questions - English

Background Questions to specify/clarify expertise

1. Could you briefly describe your professional background and your current role in quality management / standards in healthcare?

2. How long have you been working with standards in healthcare, especially in the context of hospitals and IT systems?

3. What types of standards do you work with most (e.g., quality management, information security, interoperability, clinical documentation)?

Standards-specific questions

1. Which standards do you consider most important today for handling patient data in hospitals?

2. How would you describe the current situation in German hospitals regarding compliance and implementation of these standards?

3. Where do you see the biggest discrepancies between standards "on paper" and their actual application in everyday routines?

4. What differences do you observe between large university hospitals and smaller hospitals regarding standards and standardization?

5. What concrete benefits does consistent compliance with standards offer a hospital, in your view?

6. Where do you see risks or disadvantages if standards are not or only partially complied with—especially regarding patient data?

7. How has your experience in direct patient care shaped your current view on standards and quality requirements?

8. How have you experienced the development of documentation in nursing—from paper-based to increasingly digital?

9. Which aspects of digital documentation were particularly helpful for you—and which were burdensome?

10. In stressful situations (e.g., emergencies, staff shortages, night shifts): How well did existing IT systems support you, and where did they hinder you?

11. Would you have liked to be involved in designing documentation forms or IT masks at the time? If so, what would you have changed first?

D.3 Transcripted interviews - German

Hintergrundfragen zur Spezifizierung/Klarifikation der Expertise

1. Könnten Sie kurz Ihren beruflichen Hintergrund und Ihre aktuelle Rolle im Bereich Qualitätsmanagement / Normen im Gesundheitswesen beschreiben?
   Ich bin ausgebildete Gesundheits- und Krankenpflegerin mit Berufserfahrung im Bereich Orthopädie. Anschließend habe ich einen Bachelor- und Masterabschluss im Pflege- und Gesundheitsmanagement erworben. Praktische Erfahrung im Projektmanagement habe ich im Rahmen von Digitalisierungsprojekten nach den Anforderungen des Krankenhauszukunftsgesetzes gesammelt. Seit Mai 2022 Jahren arbeite ich im Qualitätsmanagement als Stabsstelle eines Trägers mit Gesundheits- und Sozialeinrichtungen wie Krankenhäusern, Pflegeheimen und ambulanten Diensten. Mein Schwerpunkt liegt auf dem Aufbau und der

fortlaufenden Pflege der QM-Handbücher sowie der Durchführung interner Audits. Seit meinem Masterabschluss im letzten Jahr bin ich zudem als Stabsstelle Nachhaltigkeitsmanagementbeauftragte tätig und befasse mich mit den Anforderungen der CSRD und LkSG.

2. Seit wann beschäftigen Sie sich mit Normen im Gesundheitswesen, insbesondere im Kontext von Krankenhäusern und IT-Systemen? Im Rahmen meiner Tätigkeit im Qualitätsmanagement befasse ich mich seit über drei Jahren mit Normen im Gesundheitswesen unter anderem Krankenhäusern.

3. Mit welchen Arten von Normen haben Sie am meisten zu tun (z.B. Qualitätsmanagement, Informationssicherheit, Interoperabilität, klinische Dokumentation)? In erster Linie arbeite ich mit Anforderungen der ISO 9001:2015, auf deren Grundlage unser Qualitätsmanagementsystem aufgebaut ist. Diese DIN-Norm ist in all unseren Einrichtungen unabhängig vom jeweiligen Versorgungsauftrag integriert. Darauf aufbauend kommen externe Anforderungen hinzu, die der Sicherstellung der Pflegequalität dienen. In der stationären, ambulanten und teilstationären Pflege finden beispielsweise die Qualitätsprüfungs-Richtlinien (QPR) Anwendung, die vom Medizinischen Dienst (MD) und dem GKV-Spitzenverband herausgegeben werden. Dabei handelt es sich weniger um technische Normen, sondern um verpflichtende Prüfvorschriften für die unterschiedlichen Pflegebereiche. Die Einhaltung dieser Prüfkriterien wird vom MD im Rahmen externer Qualitätsprüfungen beurteilt.

Norm spezifische Fragen

1. Welche Normen sind aus Ihrer Sicht heute am wichtigsten für den Umgang mit Patientendaten in Krankenhäusern?

2. Wie würden Sie die aktuelle Situation in deutschen Krankenhäusern insichtlich der Einhaltung und Umsetzung dieser Normen beschreiben?
   Die Implementierung prozessbezogener Normen ist in deutschen Krankenhäusern in der Regel gut möglich. Schwieriger wird es allerdings, wenn es darum geht, dass die Regelungen auch tatsächlich im Alltag gelebt werden. In einem Arbeitsumfeld mit hoher Fluktuation ist es eine große Herausforderung, die Mitarbeitenden kontinuierlich in allen relevanten Normen und Standards zu schulen und zu sensibilisieren. Hier müsste

noch stärker auf Compliance und Transparenz gesetzt werden, um die Umsetzung nachhaltig zu gewährleisten.

3. Wo sehen Sie die größten Diskrepanzen zwischen „auf dem Papier“ implementierten Normen und der tatsächlichen Anwendung im Alltag? In der Praxis zeigt sich häufig, dass allein die strukturellen Voraussetzungen, also die Vielzahl an Normen und die vorhandene technische Ausstattung, nicht ausreichen. Entscheidend ist, dass diese Normen in interne Standards übersetzt und so in die alltäglichen Prozesse integriert werden, dass man tatsächlich von gelebter Compliance sprechen kann. Alle Normen und Standards nützen wenig, wenn sie nicht die starren Hierarchien in Krankenhäusern durchdringen, besonders in großen Einrichtungen wie Universitätskliniken mit komplexen Organisationsstrukturen. Nur wenn dies gelingt, können Normen echte Veränderungen bewirken und die Ergebnisqualität verbessern.

4. Inwiefern erleben Sie Unterschiede zwischen großen Universitätskliniken und kleineren Häusern in Bezug auf Normen und Standardisierung? In großen Universitätskliniken sind die Arbeitsabläufe häufig sehr getaktet, zum Beispiel im OP, und die komplexeren Eingriffe erfordern ein hohes Maß an korrekt eingehaltenen Prozessen sowie mehr personelle Kapazitäten. Außerdem gibt es dort meist zentral angesiedelte Qualitätsmanagement-Abteilungen oder Stabsstellen, die die Umsetzung von Normen steuern. Aufgrund der Vielzahl an Schnittstellen zu anderen Abteilungen und Systemen ist die Standardisierung sehr komplex. Kleinere Häuser unterliegen zwar denselben gesetzlichen Anforderungen, kommen aber aufgrund begrenzter Ressourcen oft langsamer bei der Umsetzung voran. Sie müssen zudem darauf achten, sich nicht zu stark zu überregulieren und externe Anforderungen pragmatisch handhabbar umzusetzen.

5. Welche konkreten Vorteile hat die konsequente Einhaltung von Normen für ein Krankenhaus – aus Ihrer Sicht? Die Vorteile der konsequenten Einhaltung von Normen hängen immer davon ab, aus welchem Blickwinkel man Qualität betrachtet, sei es aus Sicht der Patientinnen und Patienten, der Kostenträger oder der Leistungserbringer. Grundsätzlich geht es um das höchste Gut, die Gesundheit, bei dem nicht experimentiert werden sollte, sondern Sicherheit gewährleistet sein muss, um ein optimales Behandlungsergebnis zu erzielen. Durch die Einhaltung von

Normen lässt sich eine einheitliche Versorgungsqualität sicherstellen und Behandlungsrisiken können durch etablierte Sicherheitsbarrieren deutlich minimiert werden. Gleichzeitig ermöglichen beispielsweise Zertifizierungen, sich im Wettbewerb durch höhere Versorgungsqualität von anderen Einrichtungen abzuheben.

6. Wo sehen Sie Risiken oder Nachteile, wenn Normen nicht oder nur teilweise eingehalten werden – insbesondere beim Umgang mit Patientendaten? Wenn Normen nicht oder nur teilweise eingehalten werden, besteht die Gefahr, dass Personen die Kontrolle über ihre eigenen Daten verlieren. Hochsensible Gesundheitsdaten könnten unkontrolliert weitergegeben werden, was negative Folgen für die Patientinnen und Patienten haben kann. Gleichzeitig können solche Verstöße rechtliche Konsequenzen für die Einrichtung nach sich ziehen.

7. Wie hat Ihre Erfahrung in der direkten Patientenversorgung Ihre heutige Sicht auf Normen und Qualitätsanforderungen geprägt? Meine Erfahrung in der direkten Patientenversorgung hat meine Sicht auf Normen und Qualitätsanforderungen stark geprägt. Es geht vor allem um Verlässlichkeit: Patientinnen und Patienten verlassen sich darauf, dass sie eine optimale Versorgung erhalten, und die Pflege muss sich darauf verlassen können, dass die technische Ausstattung zuverlässig funktioniert, Daten korrekt angezeigt werden und die Prozesse möglichst wenig fehleranfällig sind. Fehler in der Patientenversorgung lassen sich nie vollständig vermeiden, deshalb ist eine offene Fehlerkultur entscheidend, Ursachenanalysen notwendig und Sicherheitslücken müssen identifiziert werden, um entsprechend handeln zu können. Gleichzeitig sind Normen oft mit bürokratischem Aufwand verbunden, gerade für bereits stark belastete Mitarbeitende. Daher ist es wichtig, dass Qualitätsmanagement von der Führungsebene aktiv gelebt wird und transparent, befähigend und handhabbar in die operative Ebene integriert wird.

8. Wie haben Sie die Entwicklung der Dokumentation in der Pflege erlebt – von eher papierbasiert zu zunehmend digital? Die Entwicklung der Dokumentation in der Pflege habe ich als sehr positiv in Bezug auf Verfügbarkeit und Zugänglichkeit erlebt. Digitale Systeme ermöglichen eine schnelle, ortsunabhängige Einsicht in Patientendaten. Gerade in interdisziplinären Teams und größeren Einrichtungen wurde die papierbasierte Dokumentation zunehmend problematisch, da sie

zeitaufwendig und fehleranfälliger ist. Neue Technologien wie Spracherkennungssysteme können dabei helfen, den Dokumentationsaufwand zu reduzieren und so Zeit für die direkte Patientenversorgung zu gewinnen.

9. Welche Aspekte der digitalen Dokumentation waren für Sie als Pflegekraft besonders hilfreich – und welche eher belastend? Besonders hilfreich an der digitalen Dokumentation war für mich die ortsunabhängige und schnelle Verfügbarkeit der Daten sowie die deutlich bessere Lesbarkeit im Vergleich zu Papierakten. Auch die Vorbereitung von Patientenakten entfiel, was den Arbeitsalltag erleichterte. Belastend waren jedoch häufige Systemupdates, während derer die Dokumentationssoftware zeitweise nicht genutzt werden konnte, sowie die Fehleranfälligkeit neuer Systeme. Insbesondere bei der Nutzung von Spracherkennungssystemen stellten sich diese als Fehleranfällig heraus, was zusätzliche Kontrollen und Zeitaufwand erforderte.

10. In Stresssituationen (z.B. Notfälle, Personalmangel, Nachtdienste): Wie gut haben die vorhandenen IT-Systeme Sie unterstützt, und wo haben sie Sie eheAusgebremst? In Stresssituationen wie Notfällen, Personalmangel oder Nachtdiensten haben die IT-Systeme vor allem dadurch unterstützt, dass Patientendaten schnell abrufbar waren, was die Entscheidungsfindung erleichtert hat. Einschränkend war jedoch, dass relevante Dokumentationen anderer Berufsgruppen häufig nur eingeschränkt greifbar waren, was die Zusammenarbeit und den Informationsfluss teilweise ausgebremst hat.

11. Hätten Sie sich damals gewünscht, bei der Gestaltung von Dokumentationsformularen oder IT-Masken mitreden zu können? Und wenn ja: Was hätten Sie als erstes geändert? Ja, ich hätte mir gewünscht, bei der Gestaltung von Dokumentationsformularen oder IT-Masken mitreden zu können. Es ist sehr wichtig, die Anwenderinnen und Anwender frühzeitig einzubeziehen und Software in der Praxis anhand von Use-Cases zu testen. Dabei würde ich besonders darauf achten, das digitale Ineinandergreifen verschiedener Systeme stärker zu nutzen und Warnhinweise zu integrieren, zum Beispiel bei der Überschreitung ärztlich angeordneter Maximaldosierungen von Medikamenten, um die Patientensicherheit zu erhöhen.

D.4 Transcripted interviews - English

Background Questions to specify/clarify expertise

1. Could you briefly describe your professional background and your current role in quality management / standards in healthcare? I am a trained nurse with professional experience in orthopedics. I later completed a bachelor's and master's degree in nursing and health management. I gained practical project management experience through digitalization projects based on the requirements of the Hospital Future Act (KHZG). Since May 2022, I have been working in quality management as a staff function for an organization operating healthcare and social facilities such as hospitals, nursing homes, and outpatient services. My main focus is building and maintaining QM manuals and conducting internal audits. Since completing my master's degree last year, I have also served as sustainability management officer, dealing with the requirements of the CSRD and the Supply Chain Due Diligence Act (LkSG).

2. How long have you been working with standards in healthcare, especially in the context of hospitals and IT systems? In my quality management role, I have been working with healthcare standards—including those in hospitals—for more than three years.

3. What types of standards do you work with most (e.g., quality management, information security, interoperability, clinical documentation)? Primarily, I work with the requirements of ISO 9001:2015, which is the foundation of our quality management system. This DIN standard is integrated into all of our facilities regardless of their care mandate. Additionally, there are external requirements ensuring quality of care. In inpatient, outpatient, and day-care nursing, for example, the Quality Inspection Guidelines (QPR) issued by the Medical Service (MD) and the GKV national association are used. These are not technical standards but binding inspection regulations for various nursing sectors. Compliance with these criteria is assessed by the MD during external quality inspections.

Standards-specific questions

1. Which standards do you consider most important today for handling patient data in hospitals? (Question was provided but not answered)

2. How would you describe the current situation in German hospitals regarding compliance and implementation of these standards? Implementing process-related standards is generally feasible in German hospitals. The real challenge is ensuring they are actually applied in daily practice. In a work environment with high staff turnover, it is extremely difficult to continuously train and sensitize employees to all relevant norms and standards. Greater emphasis on compliance and transparency is needed to ensure sustainable implementation.

3. Where do you see the biggest discrepancies between standards "on paper" and their actual application in everyday routines? In practice, structural prerequisites—such as available standards and technical equipment—are not enough. What matters is translating norms into internal standards and integrating them into everyday processes so that true compliance is achieved. Standards achieve little if they fail to permeate rigid hierarchies, especially in large institutions like university hospitals with complex organizational structures. Only if this succeeds can standards bring real change and improve outcomes.

4. What differences do you observe between large university hospitals and smaller hospitals regarding standards and standardization? In university hospitals, workflows—e.g., in the OR—are often highly structured, and complex procedures require precise process compliance and more personnel. These institutions usually have centralized QM departments guiding implementation. However, due to numerous interfaces across departments and systems, standardization becomes very complex. Smaller hospitals face the same legal requirements but implement them more slowly due to limited resources. They must also avoid overregulation and keep external requirements practical and manageable.

5. What concrete benefits does consistent compliance with standards offer a hospital, in your view? The benefits depend on whose perspective quality is viewed from—patients, insurers, or providers. Fundamentally, healthcare concerns the highest good: health. There should be no experimentation; safety must be ensured to achieve optimal treatment outcomes. Standards help ensure uniform quality of care and

significantly reduce treatment risks through established safety barriers. Certifications can also help facilities distinguish themselves through superior care quality.

6. Where do you see risks or disadvantages if standards are not or only partially complied with—especially regarding patient data? If standards are not adhered to, individuals may lose control over their own data. Highly sensitive health information may be disclosed without authorization, potentially harming patients. Such violations can also result in legal consequences for the institution.

7. How has your experience in direct patient care shaped your current view on standards and quality requirements? My clinical experience strongly influences my view. Above all, it is about reliability: patients rely on receiving optimal care, and staff rely on technology functioning properly, data being displayed correctly, and processes being as error-proof as possible. Errors can never be completely avoided, which makes an open error culture essential. Root cause analyses and identifying safety gaps are necessary. At the same time, standards often bring bureaucratic burdens, especially for already strained employees. Therefore, quality management must be actively supported by leadership and integrated transparently and practically into daily operations.

8. How have you experienced the development of documentation in nursing—from paper-based to increasingly digital? I experienced this development as very positive in terms of availability and accessibility. Digital systems allow quick, location-independent access to patient data. Especially in interdisciplinary teams and larger facilities, paper-based documentation became increasingly problematic because it was time-consuming and error-prone. New technologies such as speech recognition can help reduce documentation workload and free up time for patient care.

9. Which aspects of digital documentation were particularly helpful for you—and which were burdensome? What I found particularly helpful about digital documentation was the location-independent and fast availability of data, as well as the significantly better readability compared to paper records. The preparation of patient files was also no longer necessary, which made everyday work easier. However, it was

burdensome that frequent system updates temporarily made the documentation software unusable, and that new systems were often prone to errors. This was especially true for speech recognition systems, which proved to be error-prone and therefore required additional checking and time.

10. In stressful situations (e.g., emergencies, staff shortages, night shifts): How well did existing IT systems support you, and where did they hinder you? In stressful situations, IT systems mainly helped by allowing rapid access to patient data, facilitating decision-making. However, documentation from other professional groups was often only partially available, which slowed collaboration and information flow.

11. Would you have liked to be involved in designing documentation forms or IT masks at the time? If so, what would you have changed first? Yes, I would have liked to contribute. It is crucial to involve users early and test software in practice through use cases. I would focus on better integration between systems and implementing warning alerts—for example, when exceeding the maximum prescribed dosage of medications—to improve patient safety.

E German interview two

Interviewee: Person with working experience in numerous German hospitals as a nurse and currently employed in the field of Data security for different hospitals. As the interviewee has German as their native language, the interview was done in German and translated using ChatGPT. The questions align with the ones asked in the previous interview.

E.1 Questions - German

Hintergrundfragen zur Spezifizierung/Klarifikation der Expertise

1. Könnten Sie kurz Ihren beruflichen Hintergrund und Ihre aktuelle Rolle im Bereich Qualitätsmanagement / Normen im Gesundheitswesen beschreiben?

2. Seit wann beschäftigen Sie sich mit Normen im Gesundheitswesen, insbesondere im Kontext von Krankenhäusern und IT-Systemen?

3. Mit welchen Arten von Normen haben Sie am meisten zu tun (z.B. Qualitätsmanagement, Informationssicherheit, Interoperabilität, klinische Dokumentation)?

Norm spezifisch

1. Welche Normen sind aus Ihrer Sicht heute am wichtigsten für den Umgang mit Patientendaten in Krankenhäusern?

2. Wie würden Sie die aktuelle Situation in deutschen Krankenhäusern insichtlich der Einhaltung und Umsetzung dieser Normen beschreiben?

3. Wo sehen Sie die größten Diskrepanzen zwischen „auf dem Papier“ implementierten Normen und der tatsächlichen Anwendung im Alltag?

4. Inwiefern erleben Sie Unterschiede zwischen großen Universitätskliniken und kleineren Häusern in Bezug auf Normen und Standardisierung?

5. Welche konkreten Vorteile hat die konsequente Einhaltung von Normen für ein Krankenhaus – aus Ihrer Sicht?

6. Wo sehen Sie Risiken oder Nachteile, wenn Normen nicht oder nur teilweise eingehalten werden – insbesondere beim Umgang mit Patientendaten?

7. Wie hat Ihre Erfahrung in der direkten Patientenversorgung Ihre heutige Sicht auf Normen und Qualitätsanforderungen geprägt?

8. Wie haben Sie die Entwicklung der Dokumentation in der Pflege erlebt – von eher papierbasiert zu zunehmend digital?

9. Welche Aspekte der digitalen Dokumentation waren für Sie als Pflegekraft besonders hilfreich – und welche eher belastend?

10. In Stresssituationen (z.B. Notfälle, Personalmangel, Nachtdienste): Wie gut haben die vorhandenen IT-Systeme Sie unterstützt, und wo haben sie Sie eher Ausgebremst?

11. Hätten Sie sich damals gewünscht, bei der Gestaltung von Dokumentationsformularen oder IT-Masken mitreden zu können? Und wenn ja: Was hätten Sie als erstes geändert?

E.2 Questions - English

Background Questions to specify/clarify expertise

1. Could you briefly describe your professional background and your current role in the area of quality management / standards in healthcare?

2. Since when have you been dealing with standards in health- care, especially in the context of hospitals and IT systems?

3. Which types of standards do you deal with the most (e.g., quality management, information security, interoperability, clinical documentation)?

Standards-specific questions

1. Which standards do you consider most important today for handling patient data in hospitals?

2. How would you describe the current situation in German hospitals regarding compliance and implementation of these standards?

3. Where do you see the biggest discrepancies between standards “on paper” and their actual application in everyday routines?

4. What differences do you observe between large university hospitals and smaller hospitals regarding standards and standardization?

5. What concrete benefits does consistent compliance with standards offer a hospital, in your view?

6. Where do you see risks or disadvantages if standards are not or only partially complied with—especially regarding patient data?

7. How has your experience in direct patient care shaped your current view on standards and quality requirements?

8. How have you experienced the development of documentation in nursing—from paper-based to increasingly digital?

9. Which aspects of digital documentation were particularly helpful for you—and which were burdensome?

10. In stressful situations (e.g., emergencies, staff shortages, night shifts): How well did existing IT systems support you, and where did they hinder you?

11. Would you have liked to be involved in designing documentation forms or IT masks at the time? If so, what would you have changed first?

E.3 Transcript - German

Hintergrundfragen zur Spezifizierung/Klarifikation der Expertise Background Questions to specify/clarify expertise

1. Könnten Sie kurz Ihren beruflichen Hintergrund und Ihre aktuelle Rolle im Bereich Qualitätsmanagement / Normen im Gesundheitswesen beschreiben? Ich bin examinierte Pflegefachkraft und habe im Laufe meiner beruflichen Entwicklung verschiedene Funktionen im klinischen Bereich wahrgenommen, unter anderem als Praxisanleiter, als Stationsleitung sowie als stellvertretende Pflegedienstleitung. Parallel dazu habe ich einen Bachelor- und Masterabschluss im Gesundheits- und Pflegemanagement erworben. Heute arbeite ich als IT-Projektmanager im zentralen Dienst IT und Digitalisierung eines Krankenhauskonzerns. In dieser Funktion verantworte ich vor allem Projekte zur Einführung und Weiterentwicklung klinischer IT-Systeme, darunter KIS, LIS, PACS und RIS, sowie

die Umsetzung von Anforderungen aus Informationssicherheit, Interoperabilität und gesetzlicher Regulierung.

2. Seit wann beschäftigen Sie sich mit Normen im Gesundheitswesen, insbesondere im Kontext von Krankenhäusern und IT-Systemen?
   Die Auseinandersetzung mit Normen begleitet mich seit meinem ersten Tag in der direkten Patientenversorgung. Sowohl als Pflegekraft, als Praxisanleiter und später als Leitungskraft war die Einhaltung fachlicher, dokumentationsbezogener und qualitätsrelevanter Standards elementarer Bestandteil meiner täglichen Arbeit. Seit rund zweieinhalb Jahren beschäftige ich mich zudem intensiv mit Normen im Bereich der Krankenhaus-IT, insbesondere mit Interoperabilitäts-, Informationssicherheits- und IT-Sicherheitsanforderungen sowie den damit verbundenen regulatorischen Rahmenbedingungen.

3. Mit welchen Arten von Normen haben Sie am meisten zu tun (z.B. Qualitätsmanagement, Informationssicherheit, Interoperabilität, klinische Dokumentation)?
   Ich arbeite hauptsächlich mit folgenden Normen und gesetzlichen Vorgaben:

   - Qualitätsmanagementnormen: ISO 9001, ISO 7101, DIN 15224, DIN EN ISO 15189
   - Informationssicherheit und ISMS: ISO 27001, ISMS-Standards, Anforderungen aus NIS 1 und NIS 2
   - Datenschutz- und Patientendatenschutzgesetze, insbesondere im Kontext der elektronischen Patientenakte (ePA)
   - Interoperabilitätsstandards: FHIR, HL7
   - Medizinprodukterecht, soweit IT-Systeme als Medizinprodukte eingestuft sind

   Ein wesentlicher Teil meiner Arbeit besteht darin zu prüfen, ob klinische Systeme diesen Normen entsprechen, ob semantische und syntaktische Interoperabilität gewährleistet ist und ob die technischen, organisatorischen und prozessualen Anforderungen an Sicherheit und Dokumentation eingehalten werden.

Norm spezifisch

1. Welche Normen sind aus Ihrer Sicht heute am wichtigsten für den Umgang mit Patientendaten in Krankenhäusern?
   Der verantwortungsvolle Umgang mit Patientendaten wird aus meiner Sicht besonders durch folgende Normen und Gesetze geprägt:

   - ISO 27001 sowie ISMS-Vorgaben als Basis für Informationssicherheit
   - NIS 2, das die Anforderungen an kritische Infrastrukturen deutlich verschärft
   - ISO 9001 / DIN 15224, da prozessuale Qualität direkten Einfluss auf den Datenumgang hat
   - DIN EN ISO 15189 für diagnostische Prozesse
   - Interoperabilitätsstandards HL7 / FHIR, um Fehlübertragungen und Informationsverluste zu vermeiden
   - Datenschutzrechtliche Vorgaben, insbesondere im Zusammenhang mit cloudbasierten Systemen und internationalen Serverstandorten

2. Wie würden Sie die aktuelle Situation in deutschen Krankenhäusern insichtlich der Einhaltung und Umsetzung dieser Normen beschreiben?
   Ich halte die Einhaltung dieser Normen für insgesamt nicht ausreichend. Es kommt regelmäßig zu Datenschutzverstößen oder unsicheren Handhabungen, zum Beispiel durch:

   - geteilte Passwörter,
   - manuelle Prozessumgehungen wie gefälschte Röntgenanforderungen,
   - fehlende oder veraltete Verschlüsselung,
   - mangelhafte Benutzerrechtekonzepte.

   Viele Systeme sind zudem technologisch nicht auf dem aktuellen Stand. Schnittstellen entsprechen nicht durchgehend FHIR-Standards, wodurch Daten fehlerhaft oder unvollständig übertragen werden.

3. Wo sehen Sie die größten Diskrepanzen zwischen „auf dem Papier“ implementierten Normen und der tatsächlichen Anwendung im Alltag?
   Ich sehe deutliche Unterschiede zwischen der dokumentierten Implementierung von Normen und der tatsächlichen gelebten Praxis. Typische Beispiele:

- Sicherheitsmechanismen existieren auf dem Papier, werden aber nicht genutzt, weil Personal oder Ressourcen fehlen.
- Systeme sind technisch vorhanden, werden jedoch aufgrund unzureichender Schulung oder mangelnder Usability nicht angewendet.
- Klinische Prozesse werden trotz bestehender Standards weiterhin papierbasiert oder informell gelöst, was die Datenqualität gefährdet.

4. Inwiefern erleben Sie Unterschiede zwischen großen Universitätskliniken und kleineren Häusern in Bezug auf Normen und Standardisierung?
   Universitätskliniken verfügen über größere Fördermittel und strukturelle Ressourcen, stehen aber häufig vor einer erheblichen internen Komplexität. Entscheidungswege sind bürokratischer und langwieriger, und die Vielzahl eingesetzter Systeme führt zu hohen Integrationsanforderungen und langen Inzidenzlisten. Kleinere Häuser wiederum haben begrenztere finanzielle Mittel, können aber – wenn Führung und Strategie stimmig sind – schneller Entscheidungen treffen und umsetzen. Beide Krankenhausgrößen kämpfen jedoch mit strukturellen Defiziten in Standardisierung und Umsetzung.

5. Welche konkreten Vorteile hat die konsequente Einhaltung von Normen für ein Krankenhaus – aus Ihrer Sicht?
   Die Einhaltung von Normen bietet aus meiner Sicht mehrere zentrale Vorteile:

   - klare Regelungen für Notfall- und Krisensituationen, inklusive verantwortlicher Rollen,
   - höhere Patientensicherheit,
   - einheitliche, nachvollziehbare Prozesse,
   - reduzierte Fehlerquoten (z.B. bei Medikation oder Diagnostik),
   - verlässlichere Datenverfügbarkeit für Nachversorger,
   - höheres IT-Sicherheitsniveau und Schutz vor Datenabfluss.

6. Wo sehen Sie Risiken oder Nachteile, wenn Normen nicht oder nur teilweise eingehalten werden – insbesondere beim Umgang mit Patientendaten?
   Werden Normen nicht eingehalten, entstehen erhebliche Risiken:

- Patientendaten können in falsche Hände gelangen (z.B. durch unsichere Kommunikation oder Hackerangriffe),
- vertrauliche Informationen können missbraucht werden,
- fachliche Fehlentscheidungen können entstehen, wenn Daten falsch übertragen oder angezeigt werden,
- Medikationsfehler sind möglich, wenn Hauskataloge oder Schnittstellen nicht normgerecht arbeiten,
- rechtliche und haftungsrechtliche Konsequenzen für das Krankenhaus.

Beispiele aus meiner Erfahrung reichen von ungesicherten E-Mails bis zu fehlerhaften Laborwerten aufgrund nicht normkonformer FHIR-Schnittstellen.

7. Wie hat Ihre Erfahrung in der direkten Patientenversorgung Ihre heutige Sicht auf Normen und Qualitätsanforderungen geprägt?
   Meine Erfahrung aus der direkten Patientenversorgung hat meine heutige Sicht stark geprägt. Ich kenne sowohl die klinische Routine als auch die Belastungen in Stresssituationen. Normen sind für mich deshalb kein theoretisches Konstrukt, sondern konkrete Werkzeuge, die Patientensicherheit und Prozessqualität sichern sollen. Ich weiß, welche Folgen schlechte Dokumentation oder unstrukturierte Prozesse haben können, und warum Systeme funktional, intuitiv und sicher sein müssen.

8. Wie haben Sie die Entwicklung der Dokumentation in der Pflege erlebt – von eher papierbasiert zu zunehmend digital?
   Ich habe den Übergang von papierbasierter zu digitaler Dokumentation aus verschiedenen Perspektiven miterlebt – als Pflegekraft, als Leitung und heute als IT-Projektmanager. Durch KHZG und DiGA-Gesetze wurden viele Projekte beschleunigt, aber entscheidend ist: Man muss früh damit beginnen und die Anwender einbeziehen.

9. Welche Aspekte der digitalen Dokumentation waren für Sie als Pflegekraft besonders hilfreich – und welche eher belastend?
   Hilfreich waren für mich:

   - zentrale Verfügbarkeit aller Daten,
   - keine Weitergabe von Papierkurven mehr,
   - Nachvollziehbarkeit von Zeiten und Leistungen,

- Transparenz von Personalmangel,
- weniger Fehlerquellen in Stresssituationen.

Belastend waren:

- fehlende Workflows und nur statische Strukturbäume,
- unnötig komplexe Pflegeplanungssoftware,
- Mischformen (digital und Papier gleichzeitig), die zu Doppelarbeit führten,
- fehlende Konsistenz zwischen ärztlicher und pflegerischer Dokumentation.

10. In Stresssituationen (z.B. Notfälle, Personalmangel, Nachtdienste): Wie gut haben die vorhandenen IT-Systeme Sie unterstützt, und wo haben sie Sie eher Ausgebremst?
    In Notfällen funktionierte die digitale Dokumentation grundsätzlich gut, da alle Informationen sofort verfügbar waren und nicht erst gesucht oder herumgereicht werden mussten. Gleichzeitig bremsten fehlende digitale Prozesse aus, wenn Teile der Dokumentation weiterhin papierbasiert waren oder wenn Systeme nicht für Notfallszenarien optimiert waren.

11. Hätten Sie sich damals gewünscht, bei der Gestaltung von Dokumentationsformularen oder IT-Masken mitreden zu können? Und wenn ja: Was hätten Sie als erstes geändert?
    Ja, ich hätte mir in meiner Pflegezeit gewünscht, an der Gestaltung von Formularen und IT-Masken mitzuwirken. Heute weiß ich: Dies ist zwingend erforderlich. Nur die Fachexperten aus den jeweiligen Abteilungen kennen die relevanten Normen, Prozesse und Anforderungen. Die IT kann lediglich ein System technisch umsetzen – ob es fachlich funktioniert, entscheidet einzig die Expertise der klinischen Anwender. Ziel muss immer ein durchgängiger Workflow über alle Berufsgruppen hinweg sein – inklusive Pflege, Ärztinnen und Ärzte, Patientenverwaltung und Diagnostik.

E.4 Transcript - English

Background Questions to specify/clarify expertise

1. Could you briefly describe your professional background and your current role in the area of quality management / standards in healthcare?
   I am a registered nurse and over the course of my professional development I have held various functions in the clinical area, among others as a clinical instructor, as a ward manager, as well as as deputy director of nursing. In parallel, I obtained a Bachelor's and Master's degree in health and nursing management. Today I work as an IT project manager in the central department IT and digitalization of a hospital group. In this role, I am primarily responsible for projects for the introduction and further development of clinical IT systems, including HIS, LIS, PACS and RIS, as well as the implementation of requirements from information security, interoperability and legal regulation.

2. Since when have you been dealing with standards in healthcare, especially in the context of hospitals and IT systems?
   Dealing with standards has accompanied me since my first day in direct patient care. Both as a nurse, as a clinical instructor and later as a manager, compliance with professional, documentation-related and quality-relevant standards was an essential part of my daily work. For around two and a half years I have also been dealing intensively with standards in the area of hospital IT, in particular with interoperability, information security and IT security requirements as well as the associated regulatory framework conditions.

3. Which types of standards do you deal with the most (e.g., quality management, information security, interoperability, clinical documentation)?
   I work mainly with the following standards and legal requirements:

   - Quality management standards: ISO 9001, ISO 7101, DIN 15224, DIN EN ISO 15189
   - Information security and ISMS: ISO 27001, ISMS standards, requirements from NIS 1 and NIS 2
   - Data protection and patient data protection laws, especially in the context of the electronic patient record (ePA)
   - Interoperability standards: FHIR, HL7
   - Medical device law, insofar as IT systems are classified as medical devices

A substantial part of my work consists of checking whether clinical systems comply with these standards, whether semantic and syntactic interoperability is ensured, and whether the technical, organizational and process-related requirements for security and documentation are met.

Norms specific

1. Which standards are, in your view, the most important today for handling patient data in hospitals?
   In my view, the responsible handling of patient data is particularly shaped by the following standards and laws:

   - ISO 27001 as well as ISMS requirements as a basis for information security
   - NIS 2, which significantly tightens the requirements for critical infrastructures
   - ISO 9001 / DIN 15224, because process quality has a direct influence on data handling
   - DIN EN ISO 15189 for diagnostic processes
   - Interoperability standards HL7 / FHIR, to avoid transmission errors and loss of information
   - Data protection requirements, especially in connection with cloud-based systems and international server locations

2. How would you describe the current situation in German hospitals with regard to compliance with and implementation of these standards?
   I consider compliance with these standards overall to be not sufficient. There are regular data protection violations or insecure handling, for example through:

   - shared passwords,
   - manual process workarounds such as falsified X-ray requests,
   - missing or outdated encryption,
   - inadequate user rights concepts.

Many systems are also not technologically up to date. Interfaces do not consistently comply with FHIR standards, as a result of which data is transmitted incorrectly or incompletely.

3. Where do you see the largest discrepancies between standards implemented "on paper" and actual application in everyday work?
   I see clear differences between the documented implementation of standards and the actually lived practice. Typical examples:

   - Security mechanisms exist on paper, but are not used because staff or resources are missing.
   - Systems are technically available, but are not used due to insufficient training or lack of usability.
   - Clinical processes continue to be solved in a paper-based or informal way despite existing standards, which endangers data quality.

4. To what extent do you experience differences between large university hospitals and smaller hospitals with regard to standards and standardization?
   University hospitals have greater funding and structural resources, but often face considerable internal complexity. Decision-making paths are more bureaucratic and more lengthy, and the variety of systems used leads to high integration requirements and long incident lists. Smaller hospitals, in turn, have more limited financial resources, but can – if leadership and strategy are coherent – make decisions and implement them more quickly. However, both hospital sizes struggle with structural deficits in standardization and implementation.

5. What concrete advantages does consistent compliance with standards have for a hospital – in your view?
   In my view, compliance with standards offers several central advantages:

   - clear rules for emergency and crisis situations, including responsible roles,
   - higher patient safety,
   - uniform, traceable processes,
   - reduced error rates (e.g., in medication or diagnostics),

- more reliable data availability for follow-up care providers,
- higher IT security level and protection against data leakage.

6. Where do you see risks or disadvantages if standards are not complied with or only partially complied with – especially in dealing with patient data?
   If standards are not complied with, significant risks arise:

   - patient data can fall into the wrong hands (e.g., through insecure communication or hacker attacks),
   - confidential information can be misused,
   - professional wrong decisions can arise if data is transmitted or displayed incorrectly,
   - medication errors are possible if hospital formularies or interfaces do not work in compliance with standards,
   - legal and liability consequences for the hospital.

   Examples from my experience range from unsecured emails to incorrect laboratory values due to non-standard-compliant FHIR interfaces.

7. How has your experience in direct patient care shaped your current view of standards and quality requirements?
   My experience from direct patient care has strongly shaped my current view. I know both the clinical routine and the burdens in stress situations. For me, standards are therefore not a theoretical construct, but concrete tools that are intended to ensure patient safety and process quality. I know what consequences poor documentation or unstructured processes can have, and why systems must be functional, intuitive and secure.

8. How did you experience the development of documentation in nursing – from rather paper-based to increasingly digital?
   I experienced the transition from paper-based to digital documentation from various perspectives – as a nurse, as a manager and today as an IT project manager. Through KHZG and DiGA laws many projects were accelerated, but what is decisive is: you have to start early and involve the users.

9. Which aspects of digital documentation were particularly helpful for you as a nurse – and which were rather burdensome?
   Helpful for me were:

   - central availability of all data,
   - no more passing on of paper charts,
   - traceability of times and services,
   - transparency of staffing shortages,
   - fewer sources of error in stress situations.

   Burdensome were:

   - missing workflows and only static structure trees,
   - unnecessarily complex nursing care planning software,
   - mixed forms (digital and paper at the same time), which led to double work,
   - missing consistency between medical and nursing documentation.

10. In stress situations (e.g., emergencies, staff shortages, night shifts): How well did the existing IT systems support you, and where did they rather slow you down?
    In emergencies, digital documentation basically worked well, because all information was immediately available and did not first have to be searched for or passed around. At the same time, missing digital processes slowed things down when parts of the documentation were still paper-based or when systems were not optimized for emergency scenarios.

11. Would you have wished back then to be able to have a say in the design of documentation forms or IT masks? And if so: What would you have changed first?
    Yes, during my time in nursing I would have wished to be involved in the design of forms and IT masks. Today I know: this is absolutely necessary. Only the subject-matter experts from the respective departments know the relevant standards, processes and requirements. IT can only implement a system technically – whether it works professionally is decided solely by the expertise of the clinical users. The goal

must always be an end-to-end workflow across all professional groups – including nursing, physicians, patient administration and diagnostics.

F Romanian interview

This is the full transcript (non-, `first line`, and translated, `second new line`, versions) done via an online document sent and received by email (no human verbal interaction). The questions are an adaptation, for the current landscape of the interviewee's country, from a previous interview done by other team members. The translated version was done through Perplexity AI chatbot's capabilities.

1. Spune-ne un pic despre rolul tău în industria medicală?
   Tell us a bit about your role in the medical industry?

   1.1. Rolul tipic implică suport, administrare și integrare pentru sisteme informatice medicale (HIS, RIS, PACS, LIS), asigurarea interoperabilității, gestionarea fluxurilor clinice și administrative, precum și colaborarea cu personalul medical pentru optimizarea proceselor digitale.
   A typical role involves support, administration and integration for medical information systems (HIS, RIS, PACS, LIS), ensuring interoperability, managing clinical and administrative workflows, as well as collaborating with medical staff to optimize digital processes.

2. Cu ce sistem(e) lucrezi?
   With which system(s) do you work?

   2.1. Sunt utilizate în mod obișnuit HIS, RIS/PACS, LIS/LIMS, sisteme de raportare către CNAS (SIUI, CEAS, rețetă electronică), precum și module de programări, internări și arhivare documente.
   HIS, RIS/PACS, LIS/LIMS, reporting systems (SIUI, CEAS, electronic prescription) to Romanian's National Health Insurance House (CNAS), as well as modules for appointments, admissions and document archiving are commonly used.

   2.2. De asemenea, care sunt cele mai comune pe care le folosești?
   Also, which are the most common ones that you use?

      2.2.1. Cele mai comune sunt HIS-ul spitalului, sistemul de laborator (LIS) și sistemele de imagistică (RIS/PACS).
      The most common are the hospital HIS, the laboratory system (LIS) and the imaging systems (RIS/PACS).

3. Știi ca vreunul din standardele medicale (ex. DICOM, ICD10/11, SNOMED CT etc.) să fie aplicate în vreun sistem?
   Do you know of any medical standards (e.g. DICOM, ICD10/11, SNOMED CT etc.) that are applied in any system?

   3.1. Da, sunt utilizate standarde precum DICOM, ICD-10/11, HL7 v2/FHIR, LOINC și SNOMED CT în diverse sisteme medicale.
   Yes, standards such as DICOM, ICD-10/11, HL7 v2/FHIR, LOINC and SNOMED CT are used in various medical systems.

   3.2. Dacă da:
   If yes:

      3.2.1. Care consideri că are cel mai mare impact asupra sistemului/-sistemelor folosite?
      Which one do you consider has the greatest impact on the system(s) you use?

         3.2.1.1. DICOM are impact major asupra RIS/PACS, HL7 asupra integrării între sisteme, iar ICD-10 asupra raportării și decontării.
         DICOM has a major impact on RIS/PACS, HL7 on integration between systems, and ICD-10 on reporting and reimbursement.

         3.2.1.2. Dar asupra serviciilor medicale?
         But what about on medical services?

            3.2.1.2.1. Standardele îmbunătățesc accesul la date, reduc erorile, cresc eficiența fluxurilor clinice și asigură continuitatea îngrijirii pacientului.
            Standards improve access to data, reduce errors, increase the efficiency of clinical workflows and ensure continuity of patient care.

      3.2.2. Care sunt, din punctul tău de vedere, avantajele de a folosi aceste standarde?
      From your point of view, what are the advantages of using these standards?

         3.2.2.1. Avantajele includ interoperabilitate, date structurate, raportări corecte, reducerea erorilor, scalabilitate și integrare facilă a sistemelor noi.
         The advantages include interoperability, structured data,

correct reporting, error reduction, scalability and easy integration of new systems.

3.2.3. Poți identifica câteva probleme din pricina utilizării acestora? Dacă da, detaliază cât de mult se poate
Can you identify some problems arising from using them? If yes, describe them in as much detail as possible.

3.2.3.1. Problemele includ implementări diferite ale aceluiași standard, mesaje HL7 personalizate, completare inconsistentă a tag-urilor DICOM, codificare ICD variabilă, sisteme vechi care nu respectă standardele moderne, precum și costuri și timp ridicat pentru actualizări.
The problems include different implementations of the same standard, customized HL7 messages, inconsistent completion of DICOM tags, variable ICD coding, legacy systems that do not comply with modern standards, as well as high costs and time required for updates.

3.2.4. Există ceva standarde pe care le folosești unde lucrezi și știi să fie folosit(e) diferit în alte spitale/unități medicale?
Are there any standards that you use at your workplace which you know are used differently in other hospitals/medical units?

3.2.4.1. Da, HL7 este implementat diferit de furnizori, ICD-10 este aplicat cu rigurozitate variabilă, iar tag-urile DICOM sunt completate neuniform între unități.
Yes, HL7 is implemented differently by vendors, ICD-10 is applied with varying rigor, and DICOM tags are completed non-uniformly between facilities.

3.2.5. În general, consideri că standardele sunt o prioritate în sistemele din România?
In general, do you consider that standards are a priority in the systems in Romania?

3.2.5.1. La nivel strategic sunt o prioritate, însă în practică accentul cade adesea pe funcționalitatea imediată și pe raportările obligatorii, nu pe conformitatea strictă.
At a strategic level they are a priority, but in practice the focus often falls on immediate functionality and mandatory reporting, not on strict compliance.

4. Cunoști să existe vreun sistem de management al pacienților în care să se folosească aceste standarde?
   Do you know of any patient-management system in which these standards are used?

   4.1. Da, majoritatea HIS-urilor moderne folosesc HL7, ICD, DICOM și, în unele cazuri, LOINC/SNOMED CT.
   Yes, most modern HISs use HL7, ICD, DICOM and, in some cases, LOINC/SNOMED CT.

   4.2. Dacă da, observi potențiale riscuri sau dezavantaje dacă standardele nu sunt urmate sau doar parțial, în special când vine vorba de manevrat datele pacienților?
   If yes, do you notice potential risks or disadvantages if the standards are not followed, or followed only partially, especially when it comes to handling patient data?

      4.2.1. Riscurile includ date inconsistente, erori clinice, probleme de raportare, dificultăți de integrare și vulnerabilități în gestionarea datelor pacienților.
      Risks include inconsistent data, clinical errors, reporting issues, integration difficulties and vulnerabilities in managing patient data.

   4.3. Dacă nu, ai dori să existe ceva centralizat la nivel, măcar, regional?
   If not, would you like there to be something centralised at least at regional level?

      4.3.1. Un sistem centralizat ar reduce fragmentarea, ar îmbunătăți continuitatea îngrijirii și ar facilita schimbul de date între unități medicale.
      A centralized system would reduce fragmentation, improve continuity of care and facilitate data exchange between medical units.

5. Știi ca acesta (sistemul de mai sus) să fie integrat într-un sistem mai mare sau parte din infrastructura IT a vreunui spital/unitate medicală?
   Do you know if this system (the one above) is integrated into a larger system or part of the IT infrastructure of any hospital/medical unit?

   5.1. Da, sistemele de management al pacienților sunt integrate cu RIS/-PACS, LIS, sisteme administrative și platformele CNAS.

Yes, patient management systems are integrated with RIS/PACS, LIS, administrative systems and CNAS's platforms.

5.2. Ce beneficii consideri că aduce?
What benefits do you think this brings?

5.2.1. Beneficiile includ vedere unitară asupra pacientului, reducerea erorilor, fluxuri mai rapide, raportări automate și optimizarea resurselor medicale.
The benefits include a unified view of the patient, error reduction, faster workflows, automatic reporting and optimization of medical resources.

5.3. Există, din punctul tău de vedere, ceva neajunsuri?
Are there, in your view, any shortcomings?

5.3.1. Neajunsurile includ dependența de furnizor, complexitatea upgrade-urilor, costuri ridicate și probleme dacă integrarea este superficială.
The downsides include vendor lock-in dependency, system upgrades' complexity, high costs and problems when integration is only superficial.

6. Care crezi că ar fi punctele care iau cel mai mult timp să fie efectuate când vine vorba de sistemele IT din sistemul medical românesc?
What do you think are the tasks that take the most time to carry out when it comes to IT systems in the Romanian medical system?

6.1. Cele mai consumatoare activități sunt integrarea sistemelor, suportul pentru utilizatori, gestionarea sistemelor vechi, adaptarea la cerințe noi de raportare și testarea fluxurilor clinice.
The most time-consuming activities are system integration, user support, managing legacy systems, adapting to new reporting requirements and testing clinical workflows.

7. Cum ai vrea să fie o infrastructură de vis pentru unitățile medicale din țară?
How would you like an ideal infrastructure for medical units in the country to look?

7.1. O infrastructură ideală ar include identificator unic național, HIS modern și modular, interoperabilitate completă (HL7 FHIR, DICOM, ICD-11, SNOMED CT), integrare nativă RIS/PACS/LIS,

platformă națională de schimb de date, securitate avansată, audit complet, training continuu și infrastructură hardware modernă și redundantă.
An ideal infrastructure would include a national unique identifier, a modern and modular HIS, full interoperability (HL7 FHIR, DICOM, ICD-11, SNOMED CT), native integration of RIS/PACS/LIS, a national data-exchange platform, advanced security, full audit, continuous training and modern, redundant hardware infrastructure.

G Interview consent form



Interviewee Information			
Name			
D.O.B.		Occupation	

Interview purpose: I understand that I am participating in a research Interview and that the data collected will be used for academic purposes.

	Yes	No
I do consent to be recorded	☐	☐
I do consent to an anonymous publication of the interview	☐	☐

Date _____ Signature _____

Figure 7